

Experiment 18 :

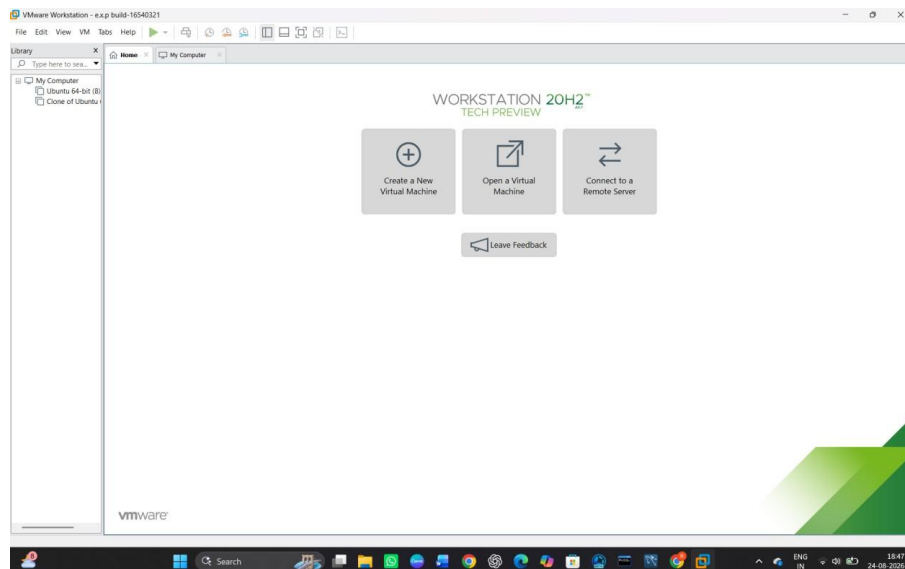
Install Virtualbox/VMware Workstation with different flavours of linux or windows OS on top of your present windows environment.

AIM:

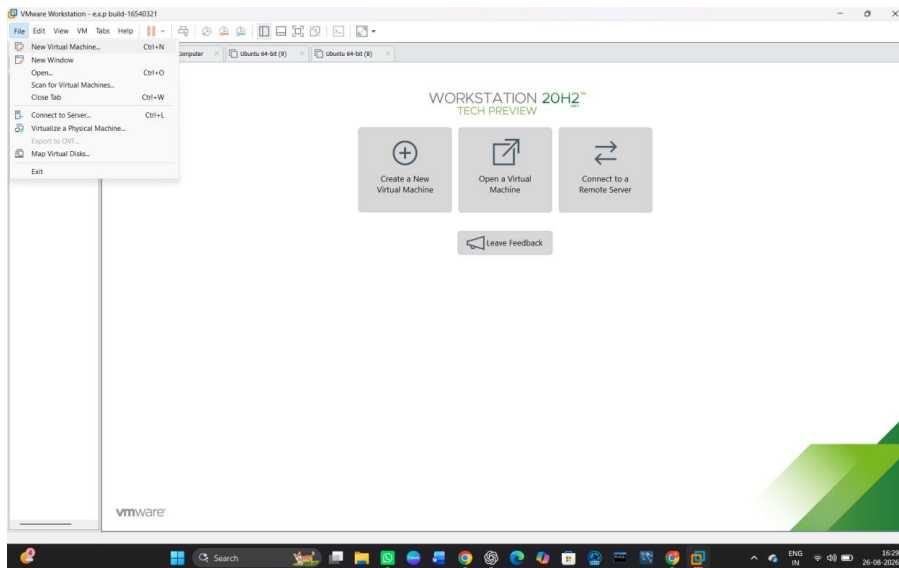
To install VMware Workstation and install a different flavour of Linux/Windows Operating System as a virtual machine on the existing Windows environment.

PROCEDURE:

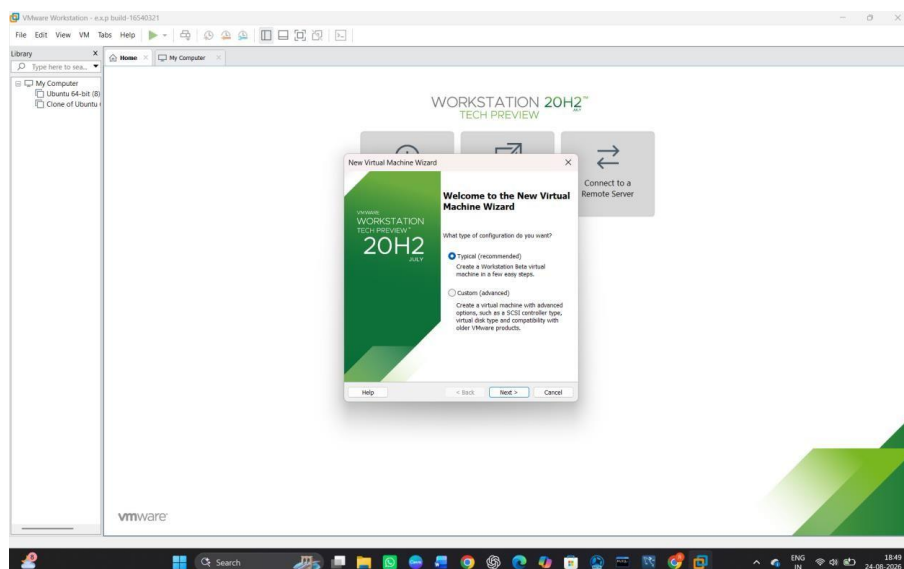
STEP 1: Go to VMware Workstation.



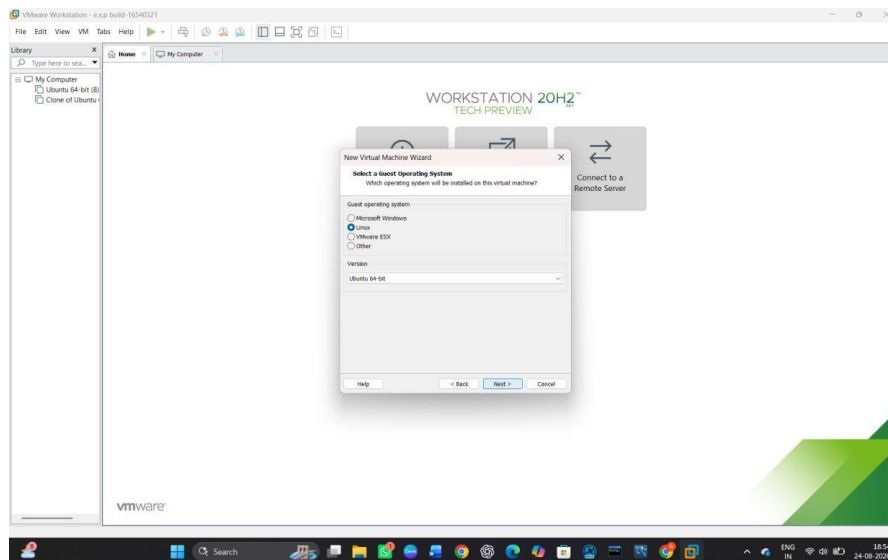
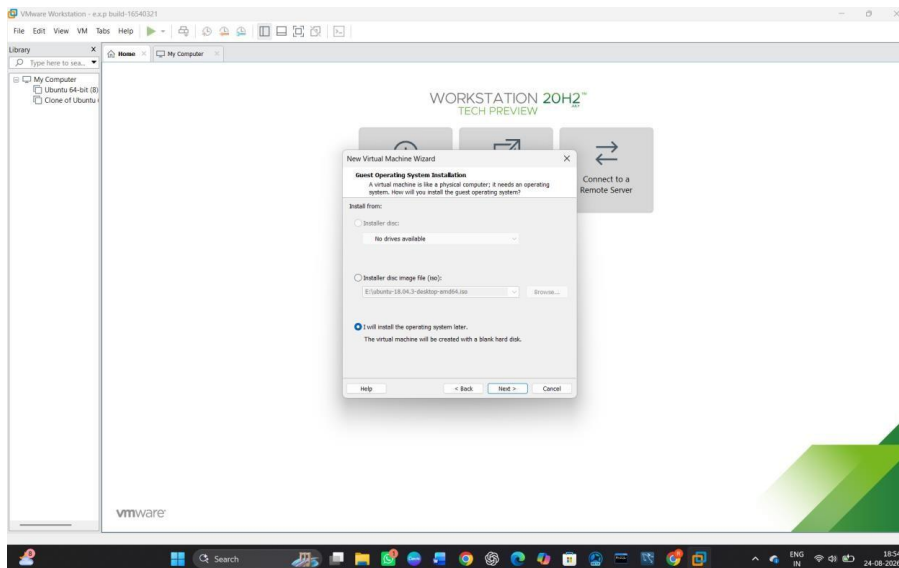
STEP 2: Click on Create a New Virtual Machine.



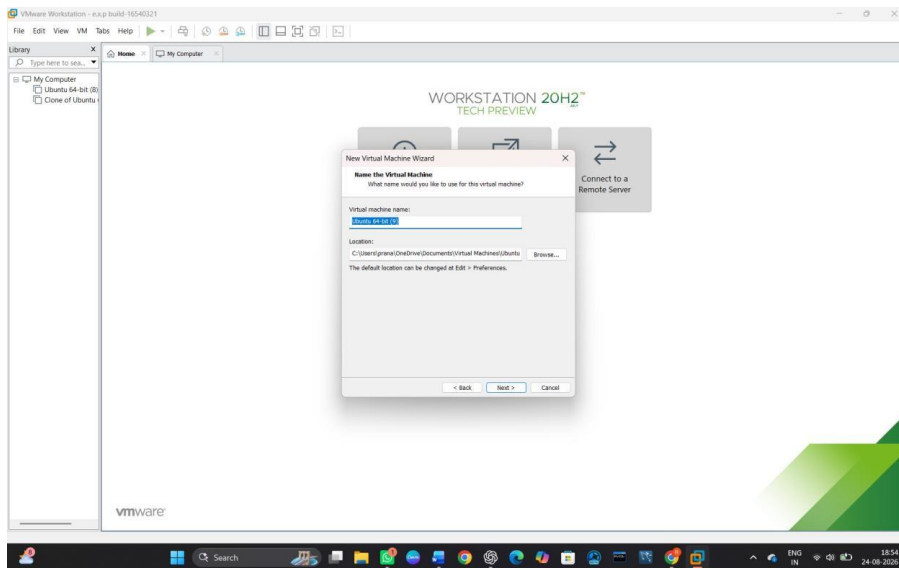
STEP 3: Select Typical (Recommended) and click Next.



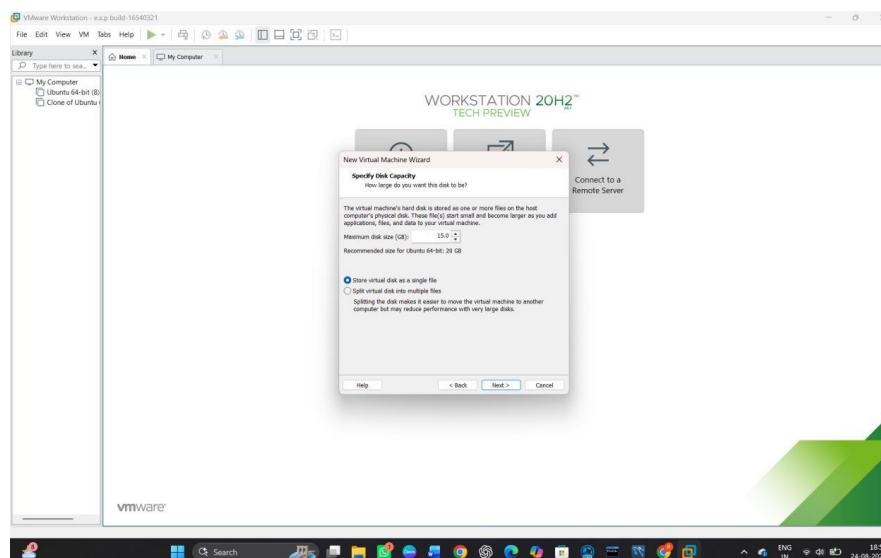
STEP 4: Select the ISO image file of the Linux/Windows operating system.



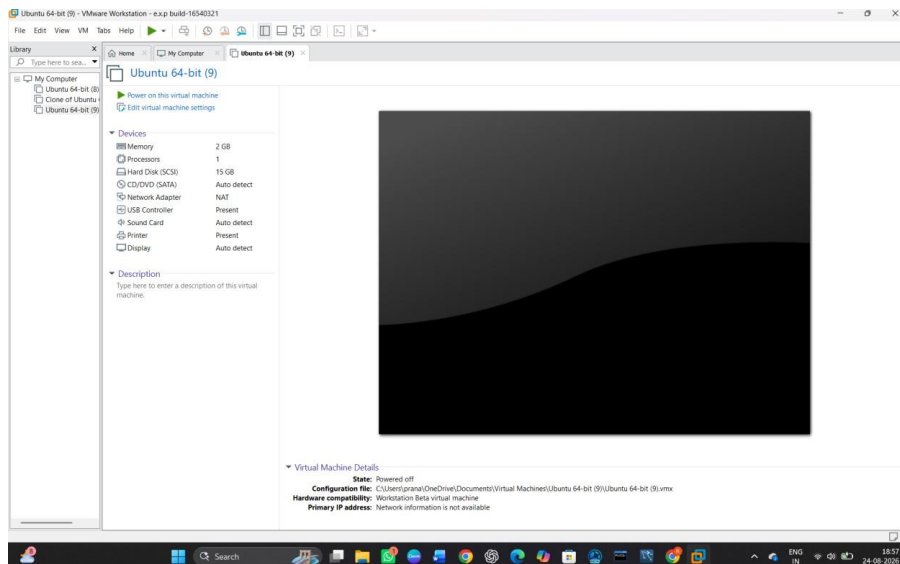
STEP 5: Enter the Virtual Machine Name and select the storage location.



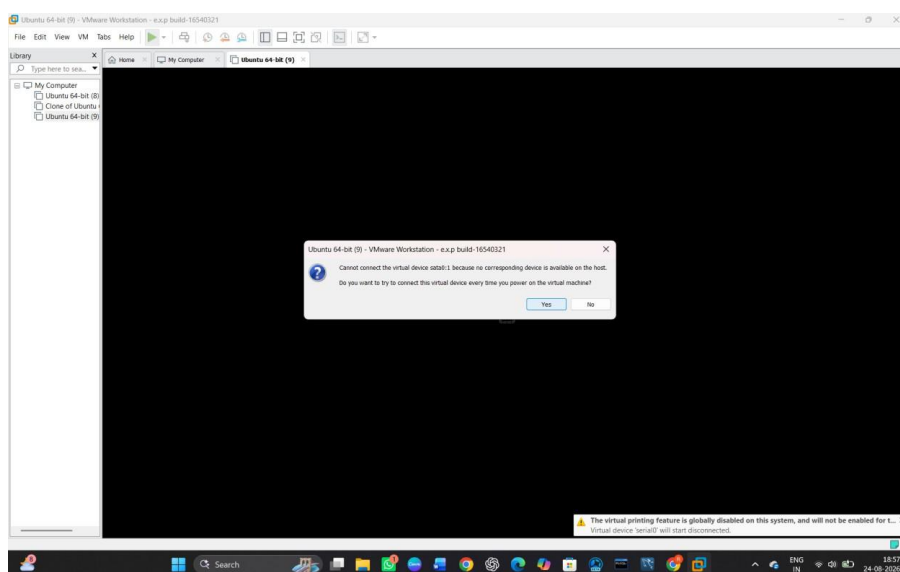
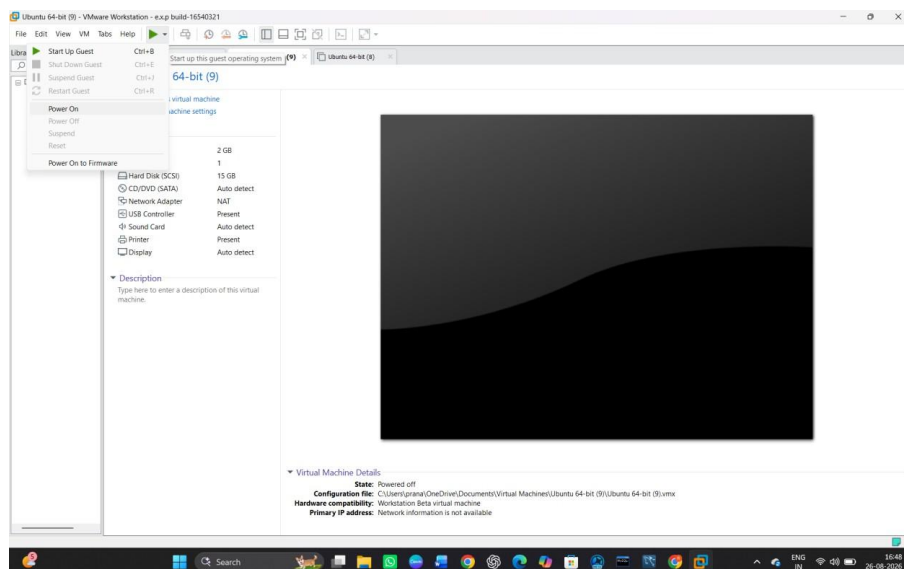
STEP 6: Specify the required disk size and click Next.



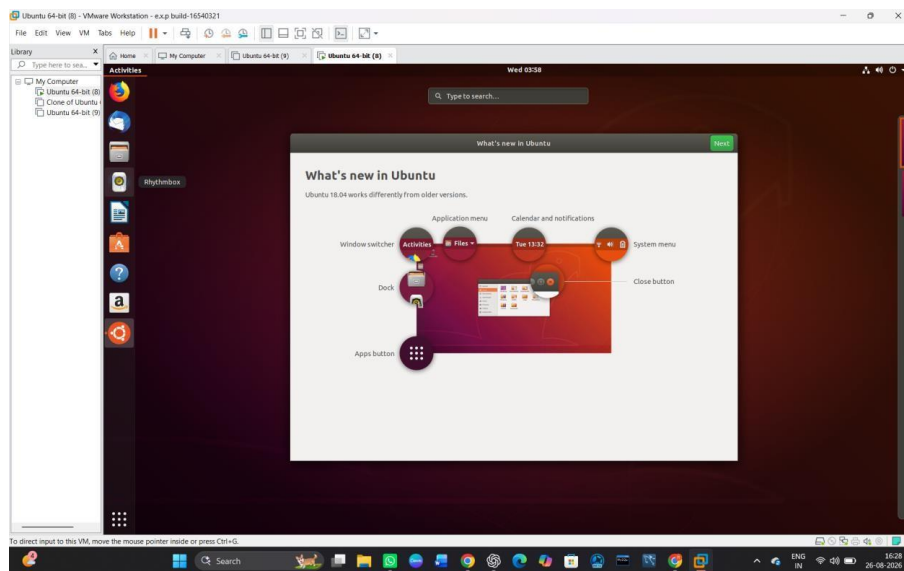
STEP 7: Click on Customize Hardware and configure the required RAM and processor.



STEP 8: Click Power on this virtual machine.



STEP 9: Install the selected Linux/Windows operating system by following the installation instructions.



Experiment 19:

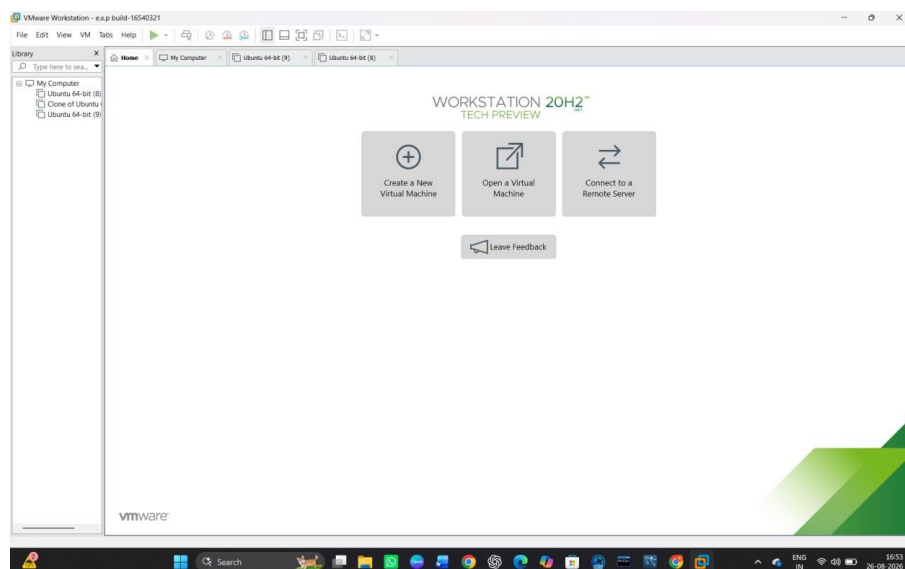
Install a C compiler in the virtual machine created using virtual box / VM Ware Workstation / Player and execute Simple C Programs.

AIM:

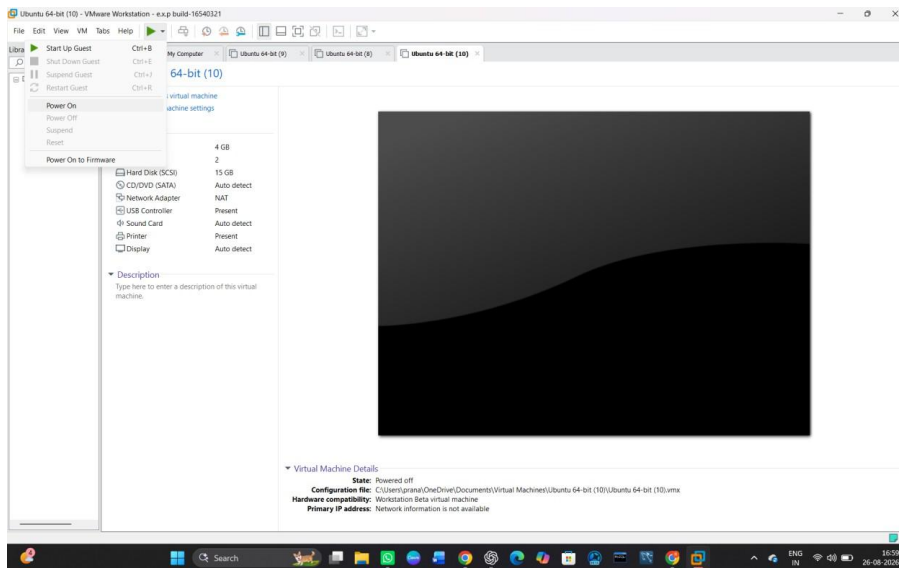
To install a C compiler (GCC) in the virtual machine created using VMware Workstation and execute simple C programs.

PROCEDURE:

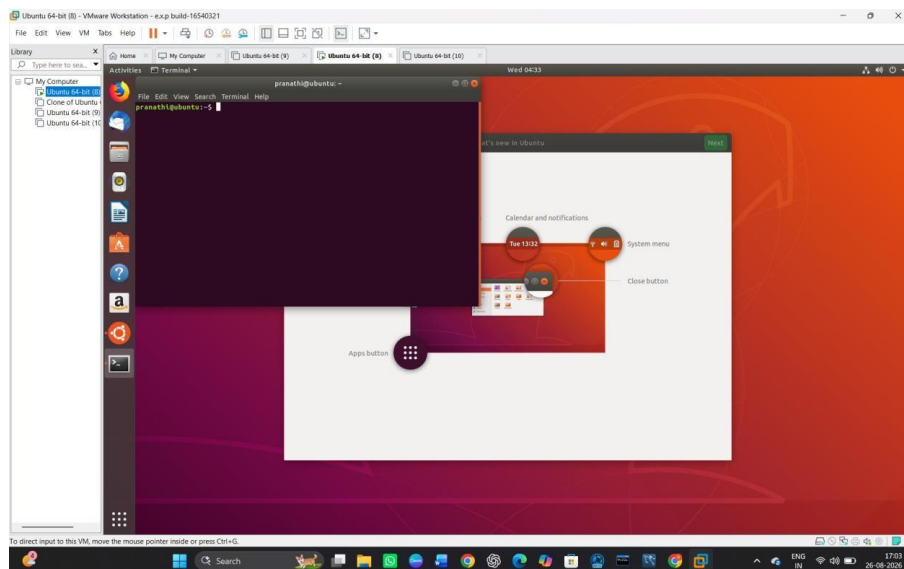
STEP 1: Go to VMware Workstation.



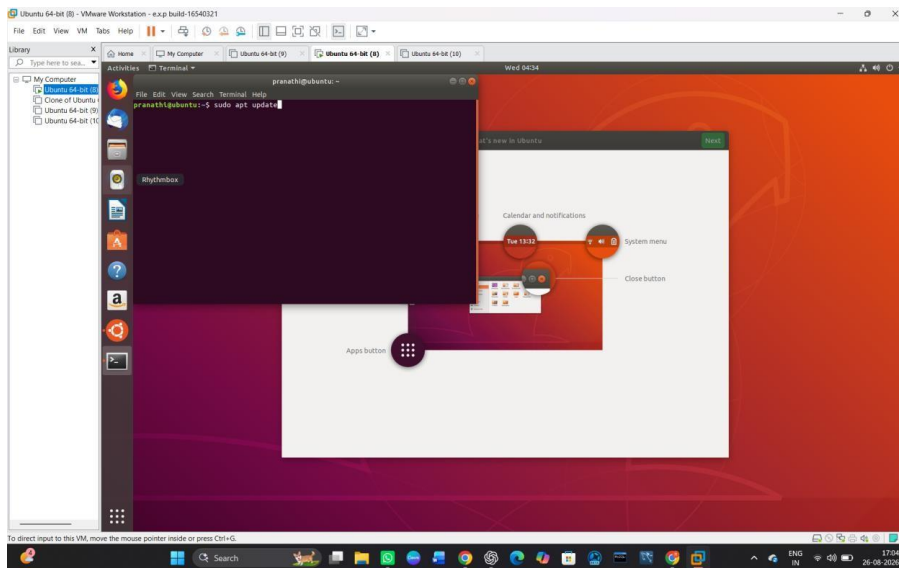
STEP 2: Start the Ubuntu virtual machine.



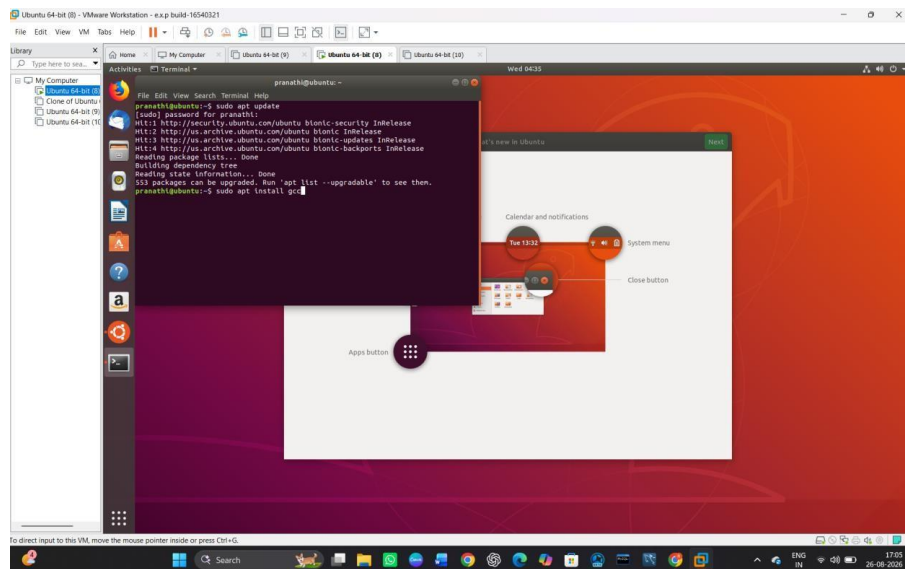
STEP 3: Open the Terminal in Ubuntu.



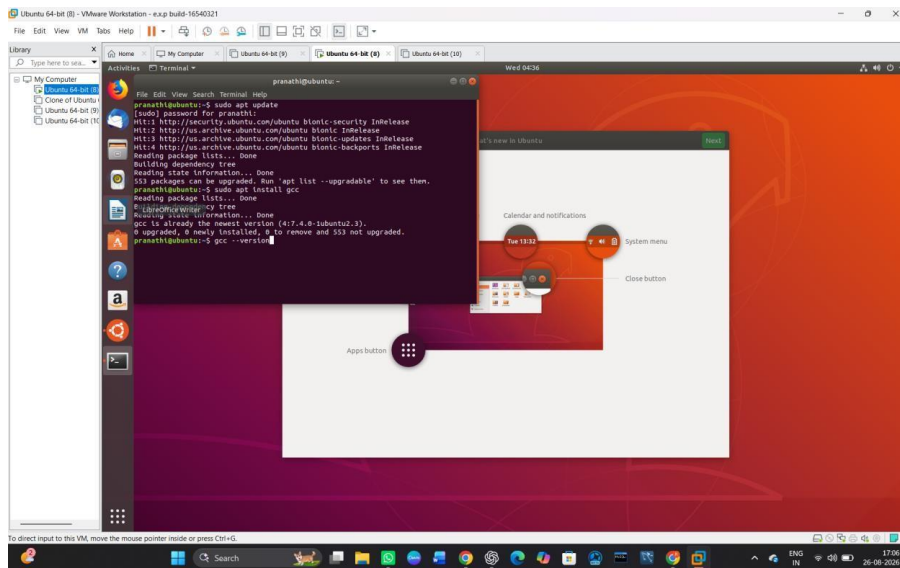
STEP 4: Update the package list using the command:



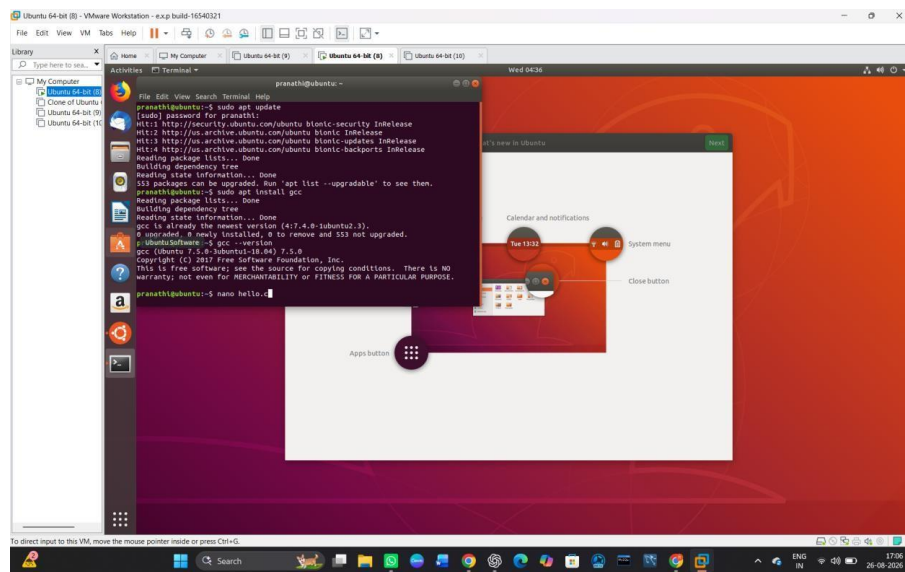
STEP 5: Install the GCC compiler using:



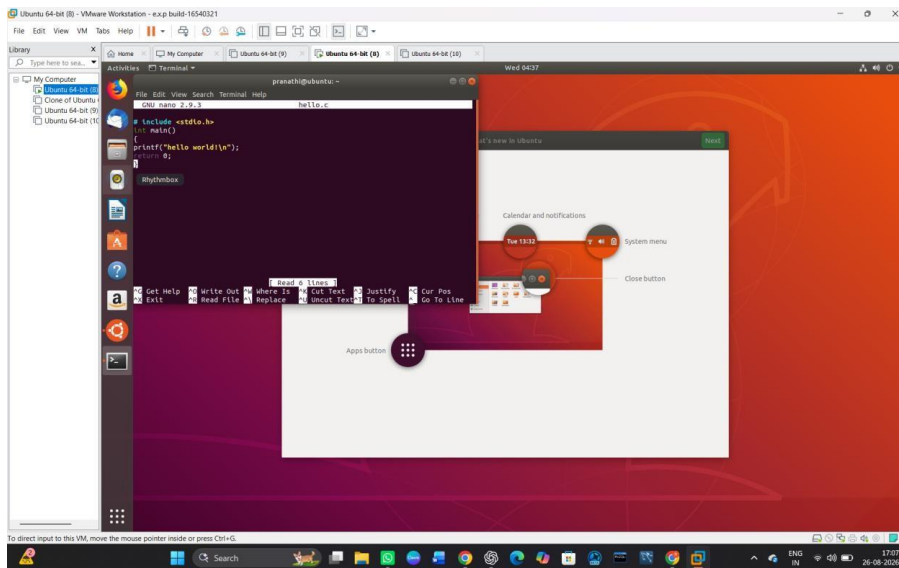
STEP 6: Check whether GCC is installed using:



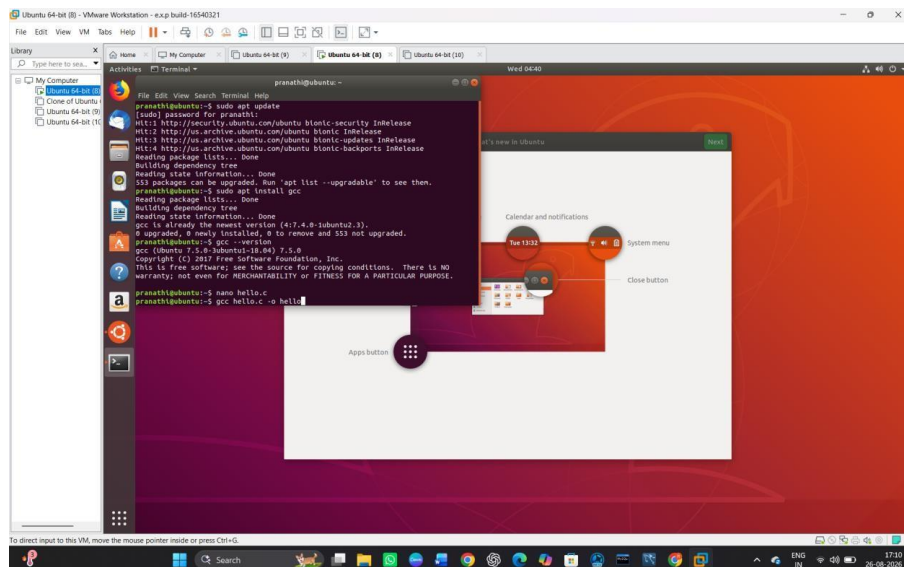
STEP 7: Create a C program using:



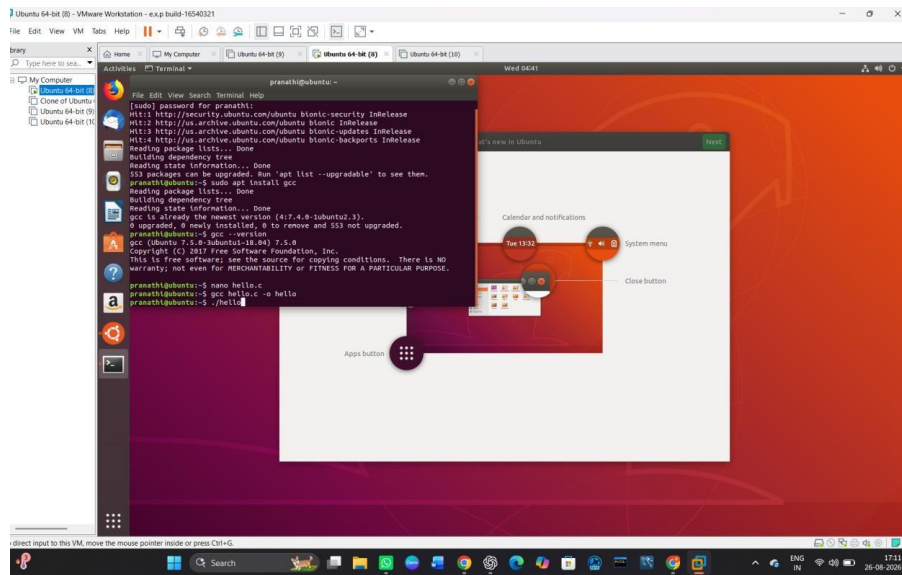
STEP 8: Enter a simple C program and save the file.



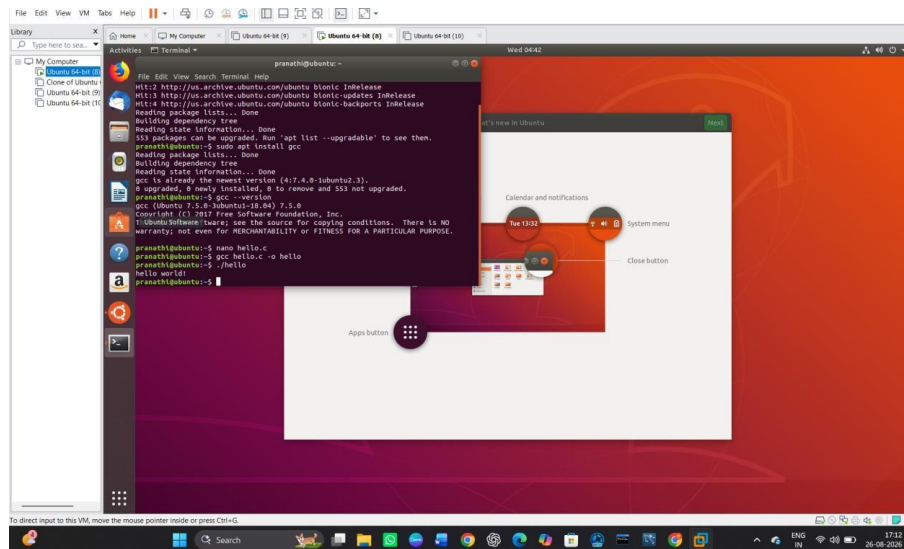
STEP 9: Compile the program using:



STEP 10: Execute the program using:



STEP 11: Successfully Executed C program



Experiment 20 :

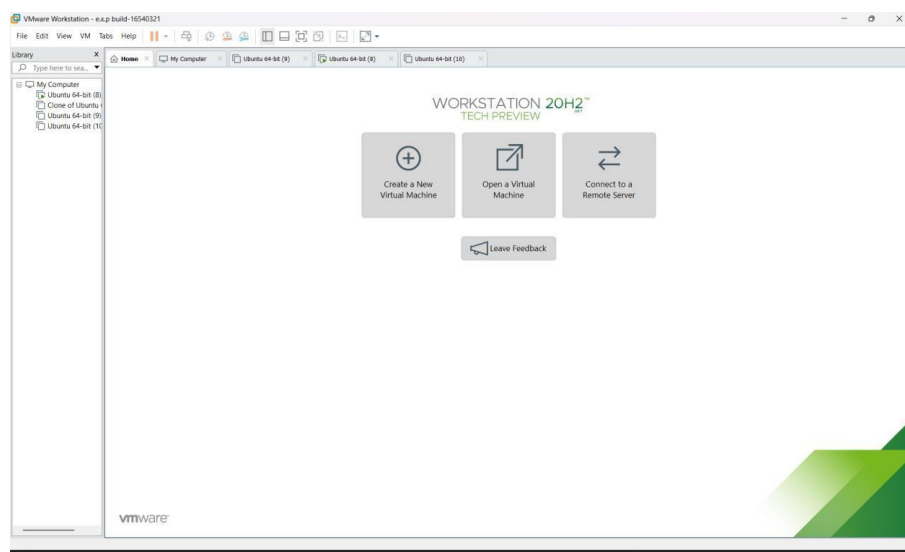
Install Virtualbox with different flavours of linux or windows OS on top of windows 10 using custom installation.

AIM:

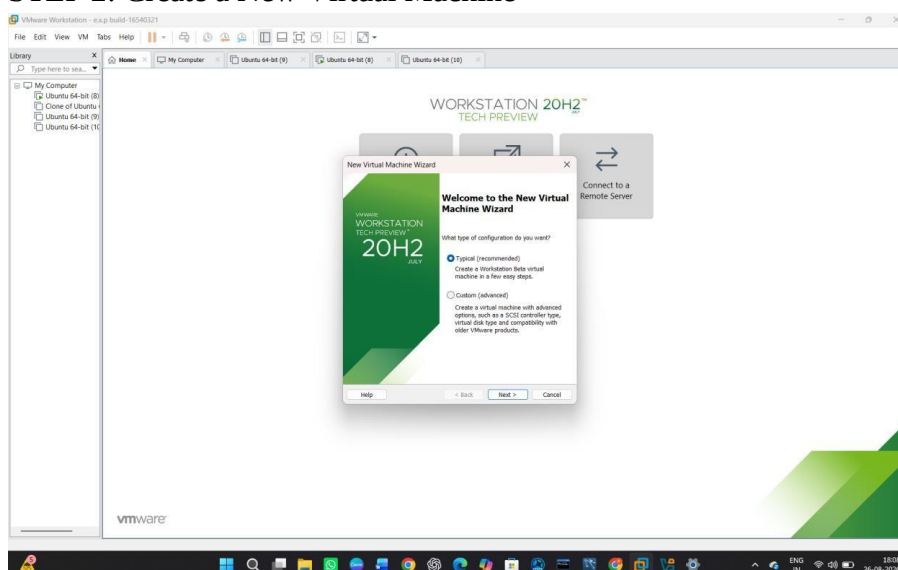
To install Oracle VirtualBox with a different flavour of Linux/Windows Operating System on top of the existing Windows 10 environment using Custom Installation.

PROCEDURE:

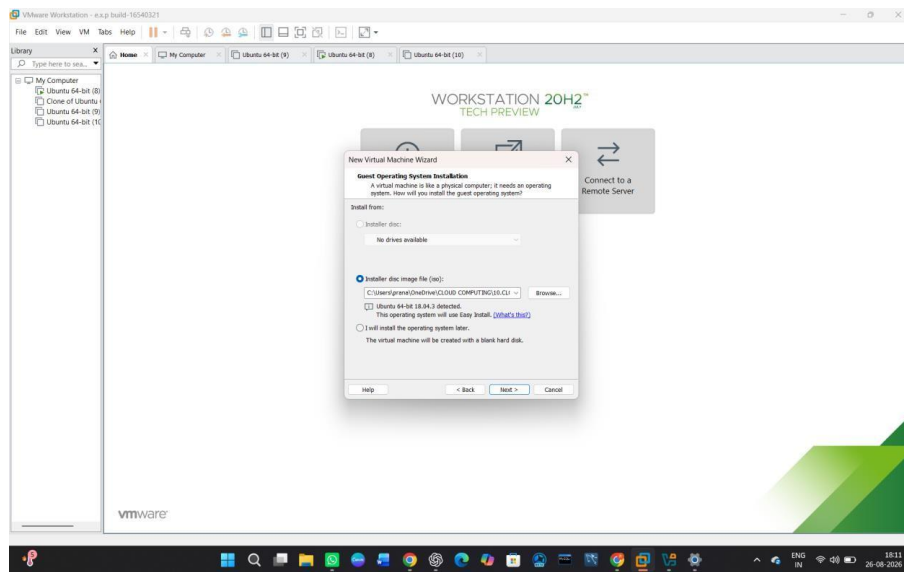
STEP 1: Open VMware Workstation



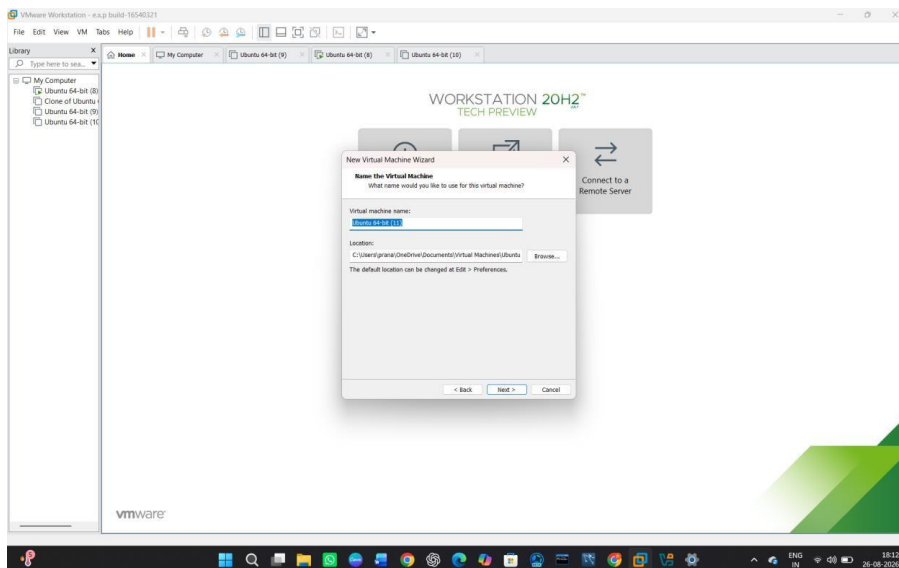
STEP 2: Create a New Virtual Machine



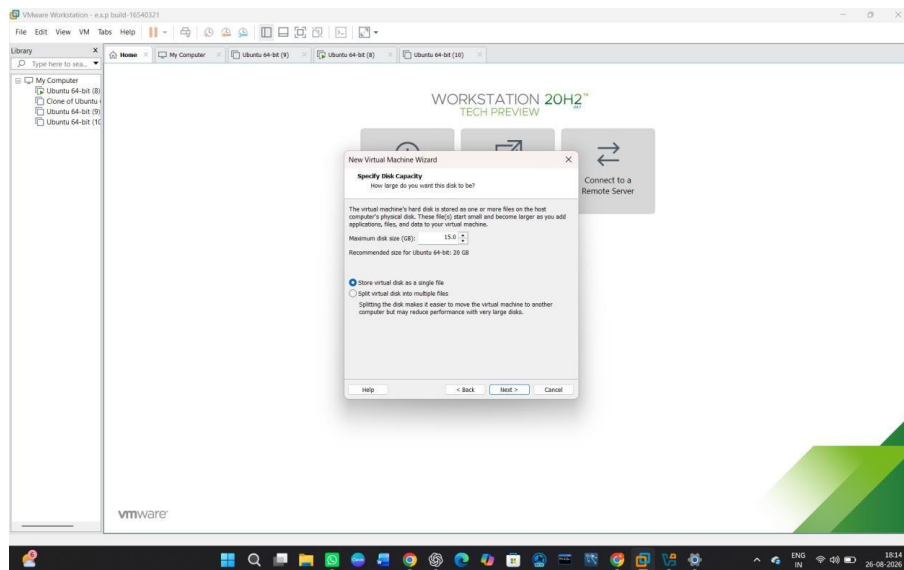
STEP 3: Select Ubuntu ISO



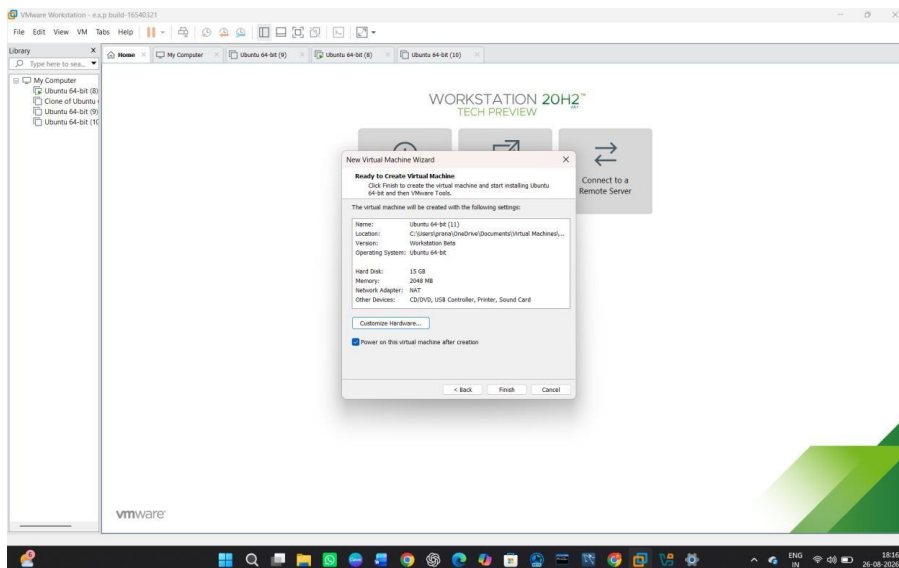
STEP 4: Give the VM a Name



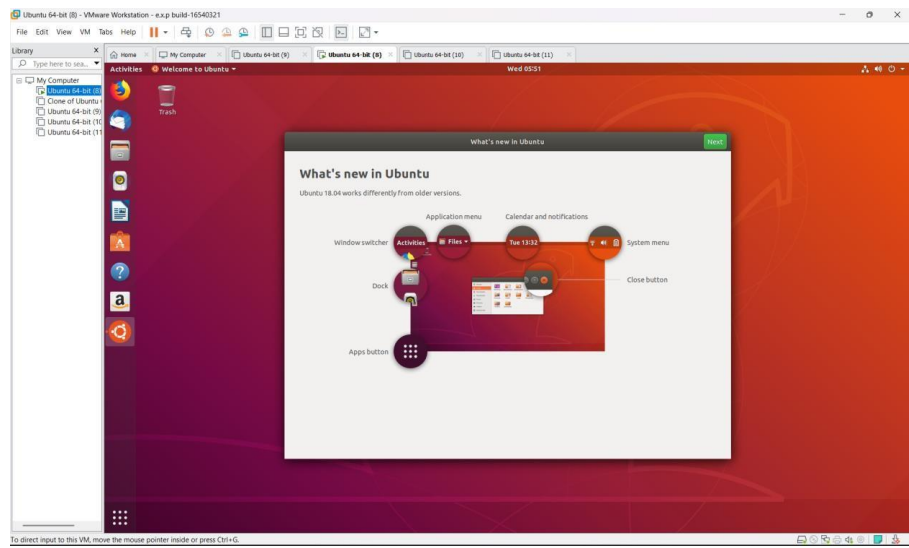
STEP 5: Configure the Virtual Disk



STEP 6: Customize Hardware



STEP 7: Start the Virtual Machine



Experiment 21 :

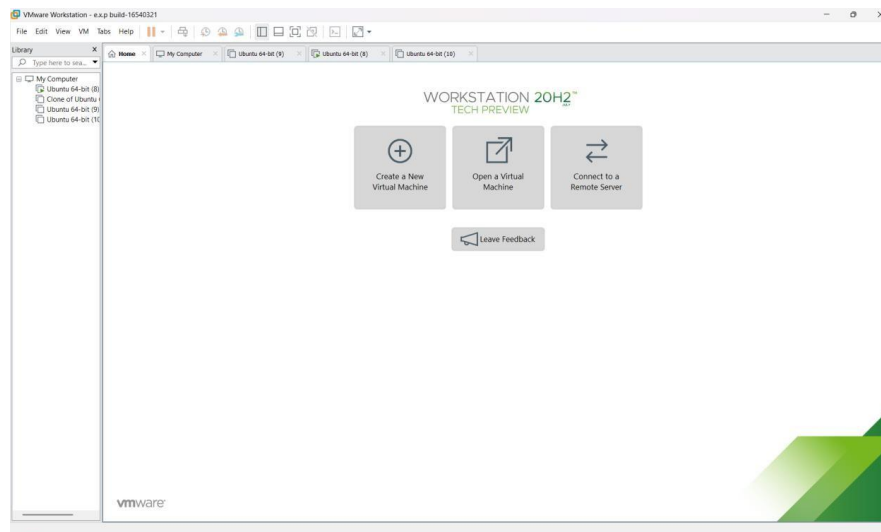
Install Virtualbox/VMware Workstation with different flavours of linux or windows OS on top of windows 7 or 8.

AIM:

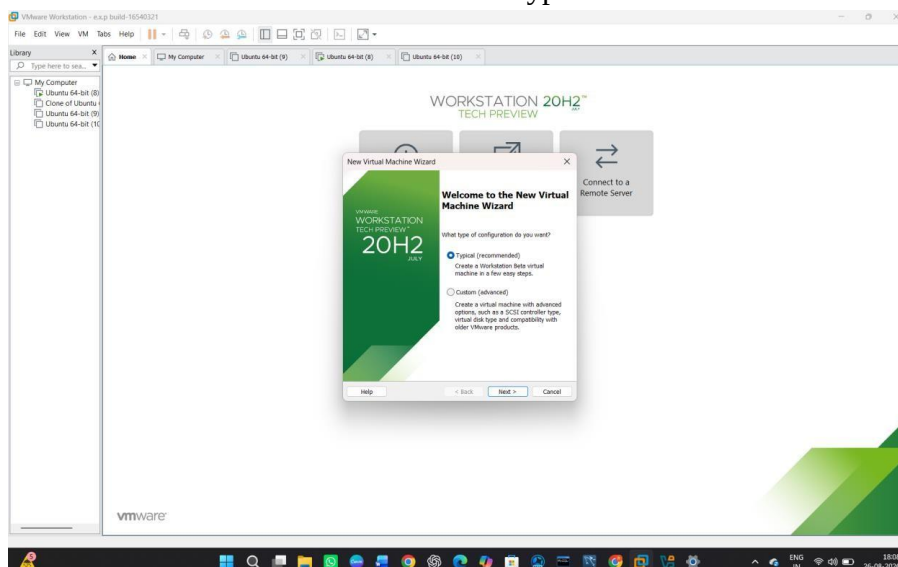
To install VMware Workstation and install a different flavour of Linux/Windows Operating System on top of the existing Windows 7/8 host environment.

PROCEDURE:

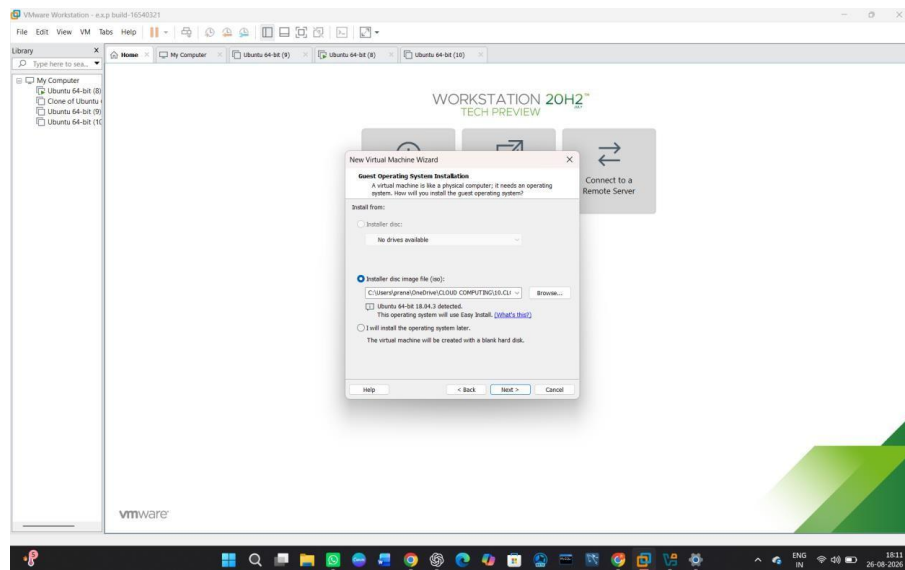
STEP 1: Go to VMware Workstation.



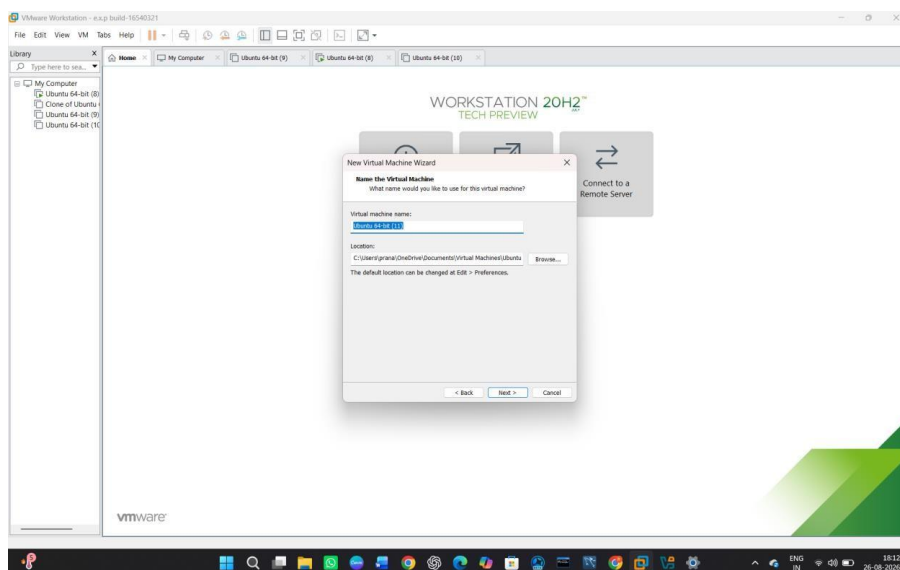
STEP 2 — Select the Virtual Machine Type



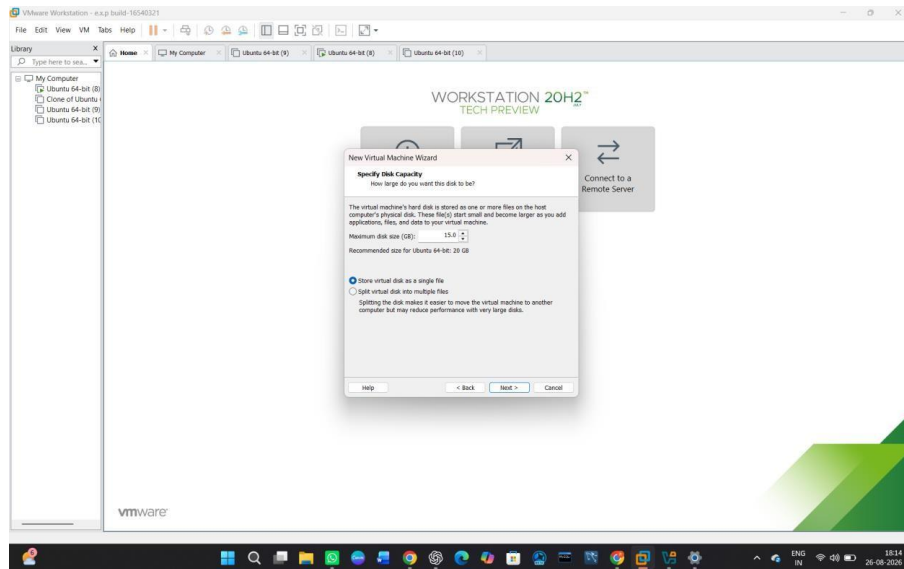
STEP 3 — Select the Ubuntu ISO



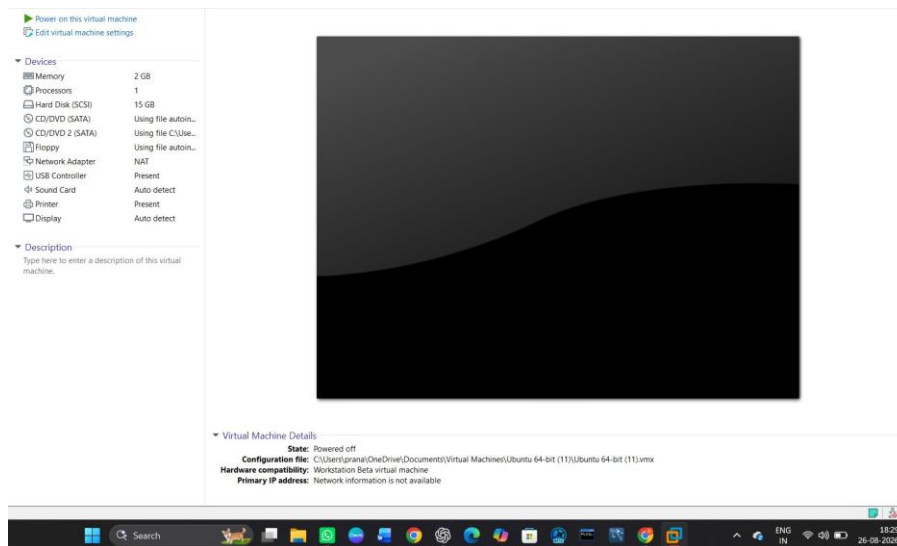
STEP 4 — Name the Virtual Machine



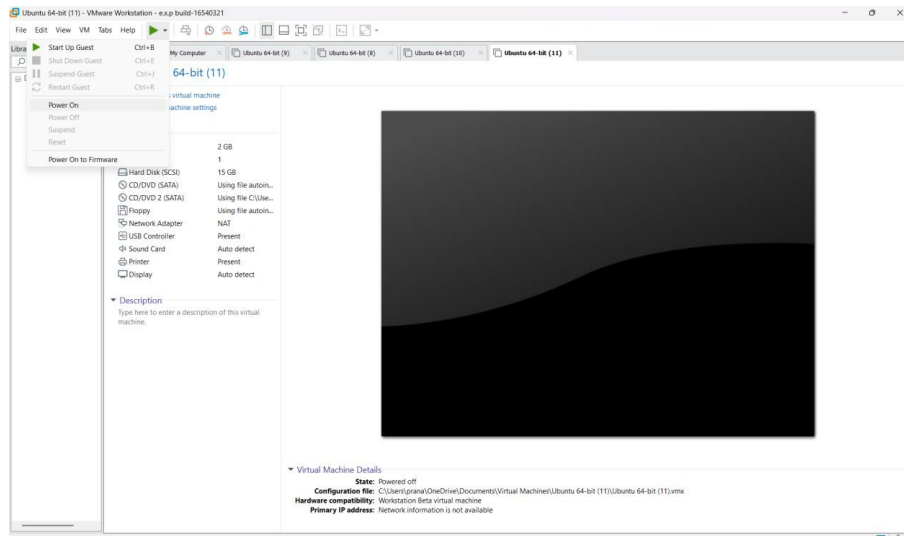
STEP 5 — Configure Disk Capacity



STEP 6 — Customize Hardware



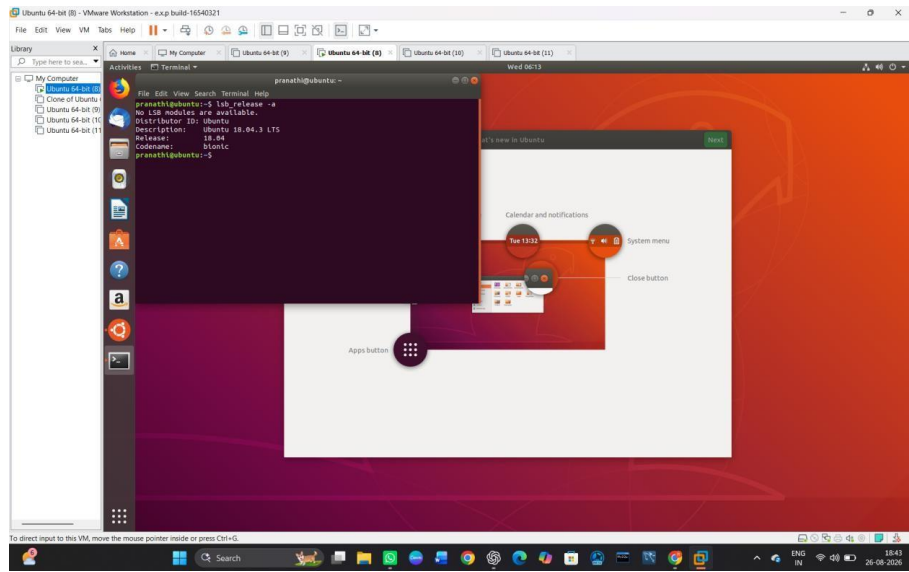
STEP 7 — Power On the Virtual Machine



STEP 8 — Install Ubuntu and Login



STEP 9 — Verify the Installation



Experiment 22 :

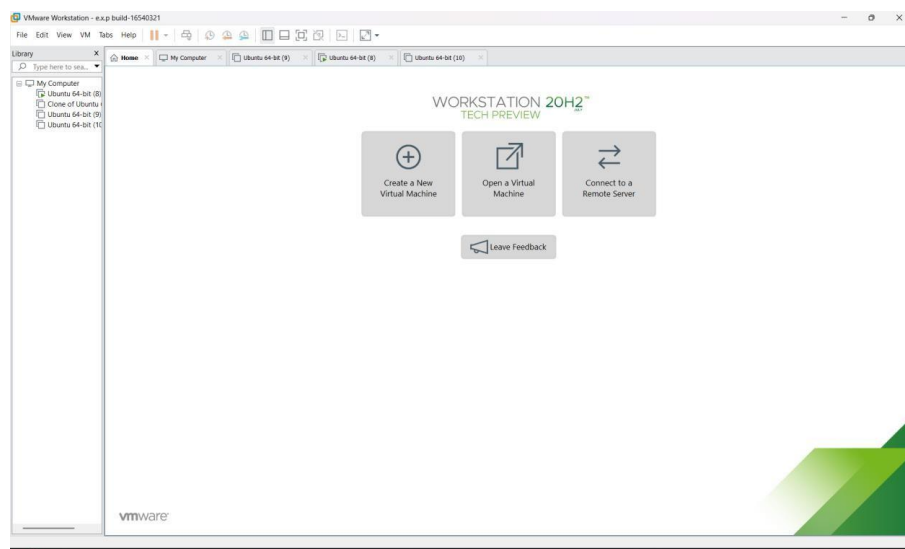
To write a procedure to run the virtual machine of different configuration and to check how many virtual machines can be utilized at a particular time.

AIM:

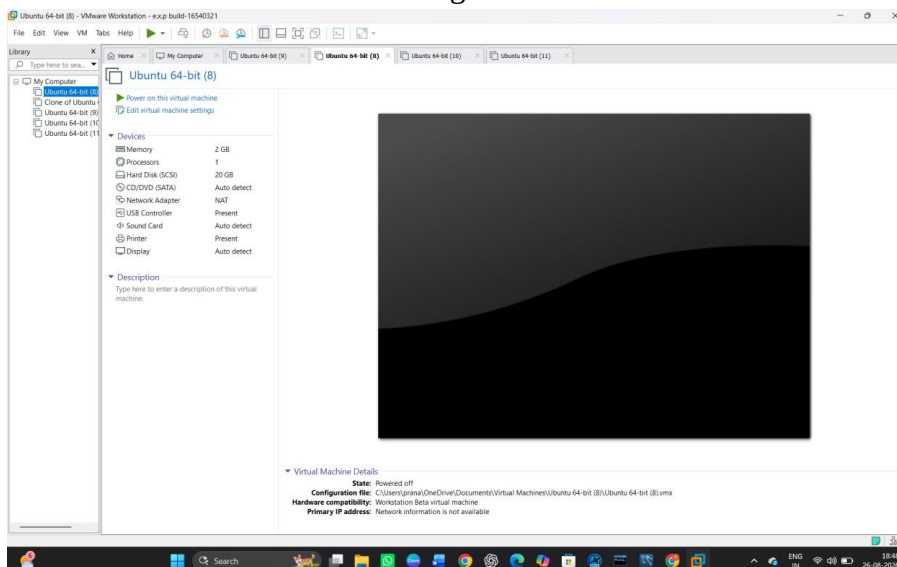
To run virtual machines with different configurations and determine how many virtual machines can be utilized simultaneously on a host system.

PROCEDURE:

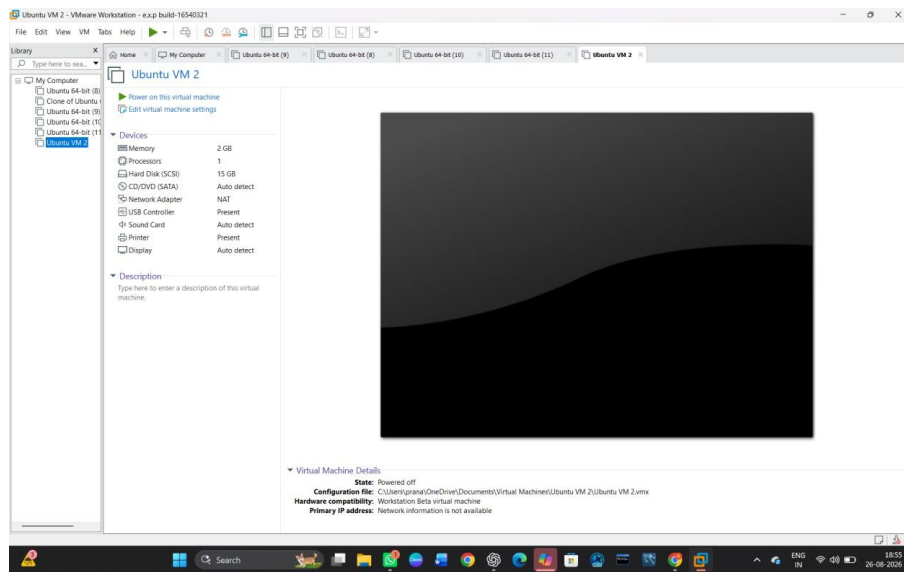
STEP 1— Go to VMware Workstation.



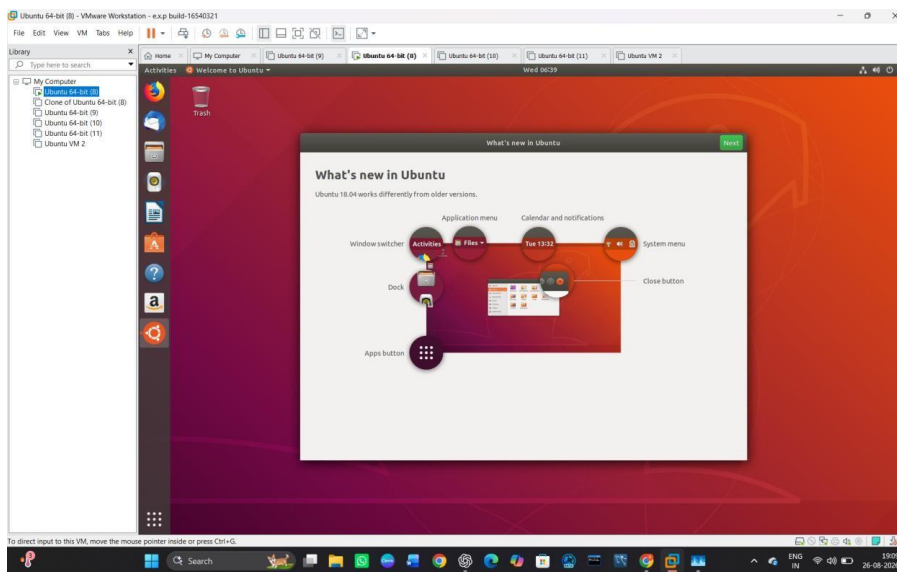
STEP 2 — Check Your VM Configuration



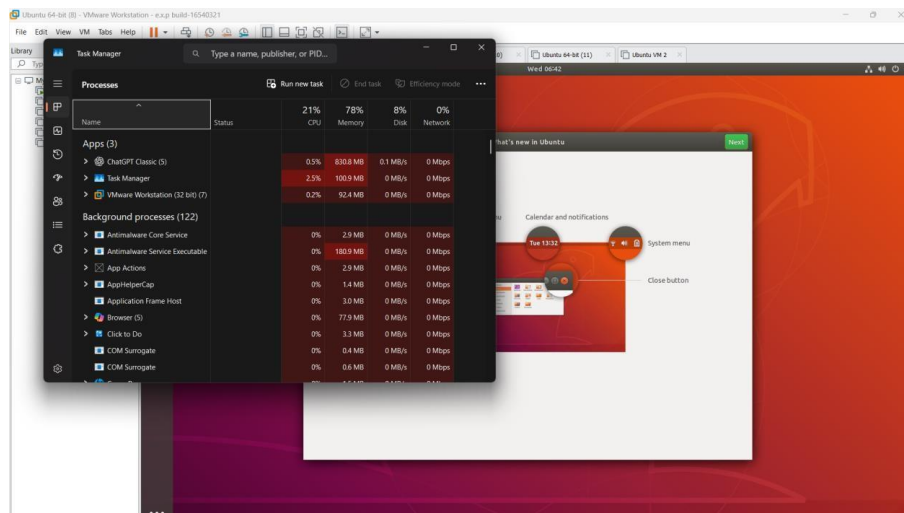
STEP 3 — Create the Second Virtual Machine



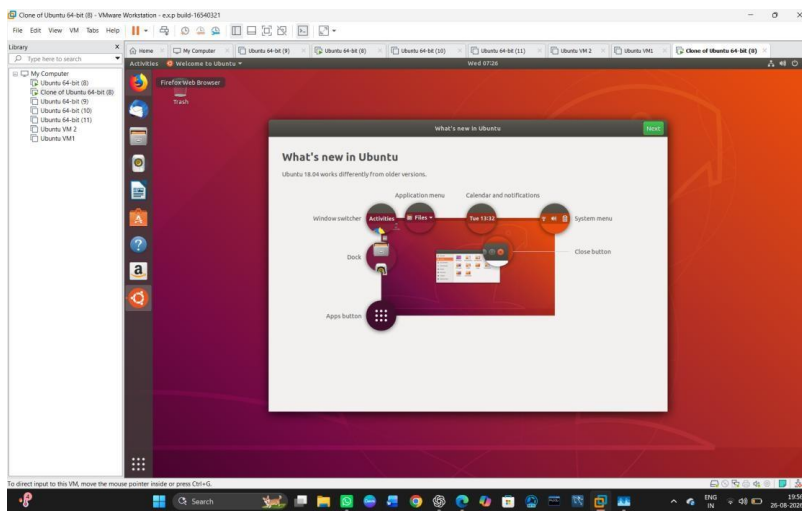
STEP 4 — Start the First VM



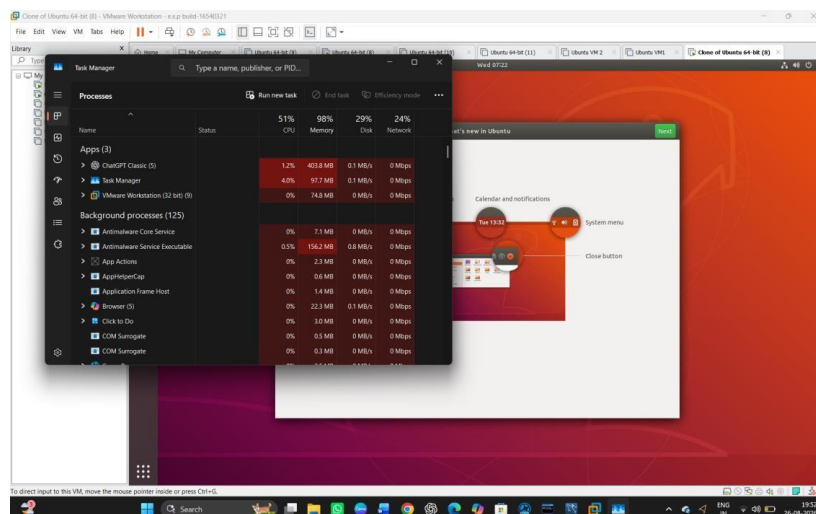
STEP 5 — Check the Host Resources



STEP 6 — Start the Second VM



STEP 7 — Check the Second VM



Result

Two virtual machines were successfully run simultaneously using VMware Workstation. The CPU and memory utilization were monitored using Windows Task Manager. When two VMs were running, memory utilization reached 98%, showing that the available system resources limited the number of VMs that could be utilized simultaneously.

Experiment 23 :

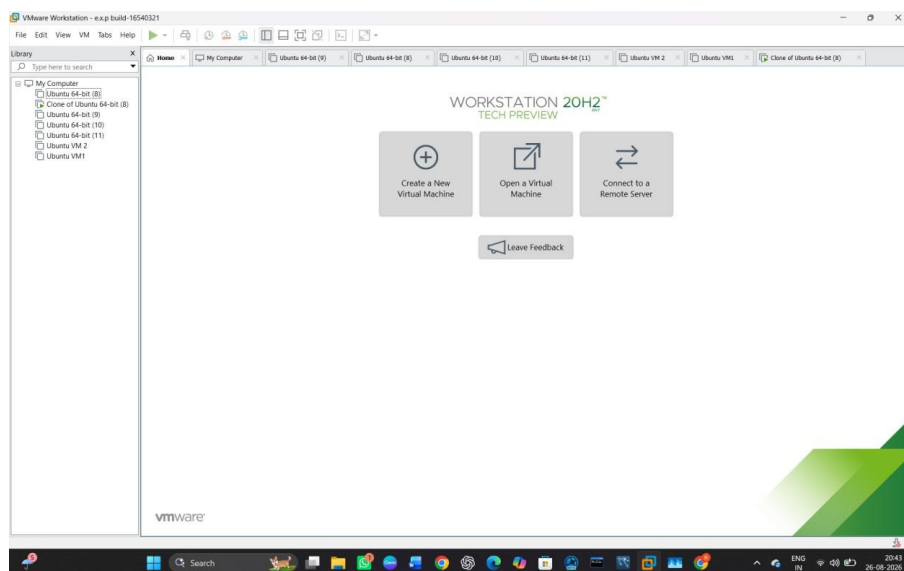
Write a Procedure To Attach Virtual Block To The Virtual Machine And Check Whether It Holds the Data Even After The Release Of The Virtual Machine

AIM:

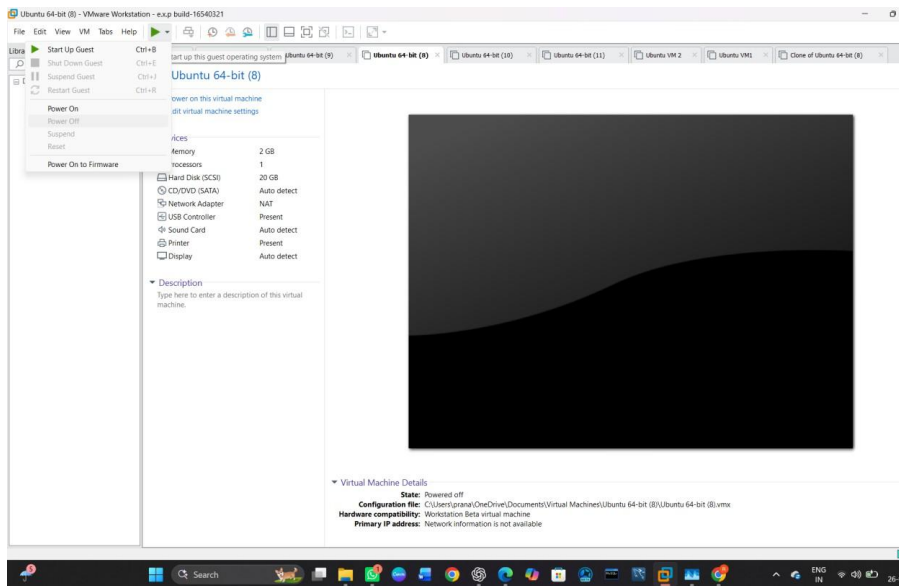
To attach a virtual block storage disk to a virtual machine, store data in it, release/detach the virtual machine, and verify that the data is retained.

PROCEDURE:

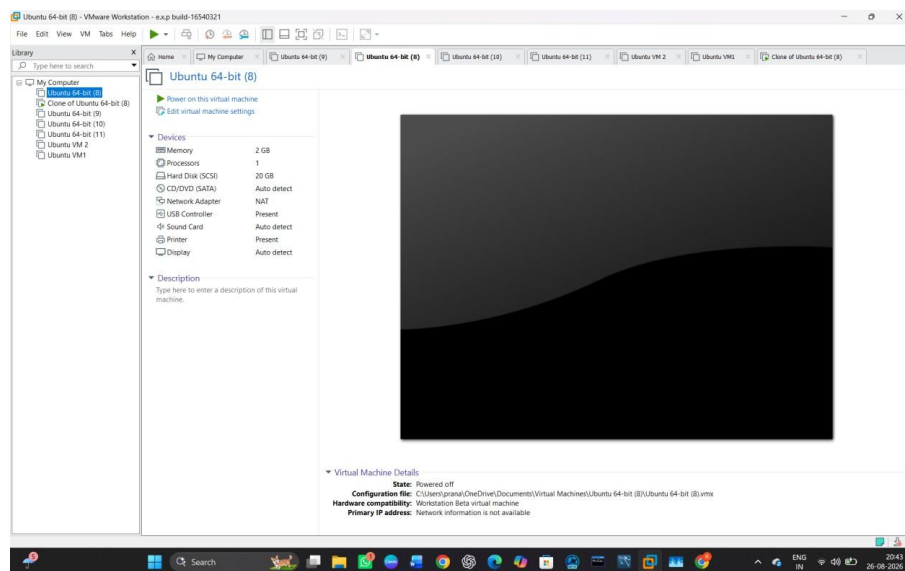
STEP 1: Open VMware Workstation.



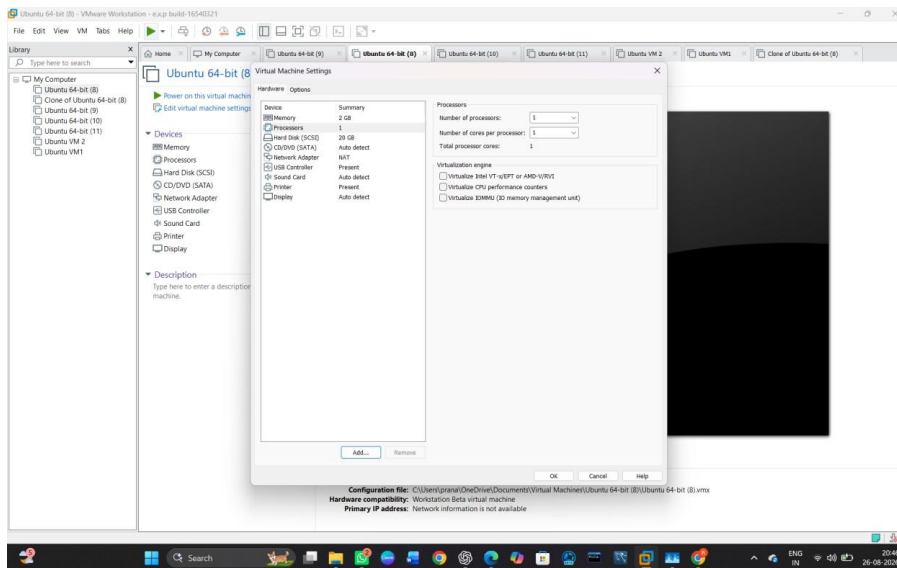
STEP 2: Select the Ubuntu Virtual Machine and Power Off Ubuntu



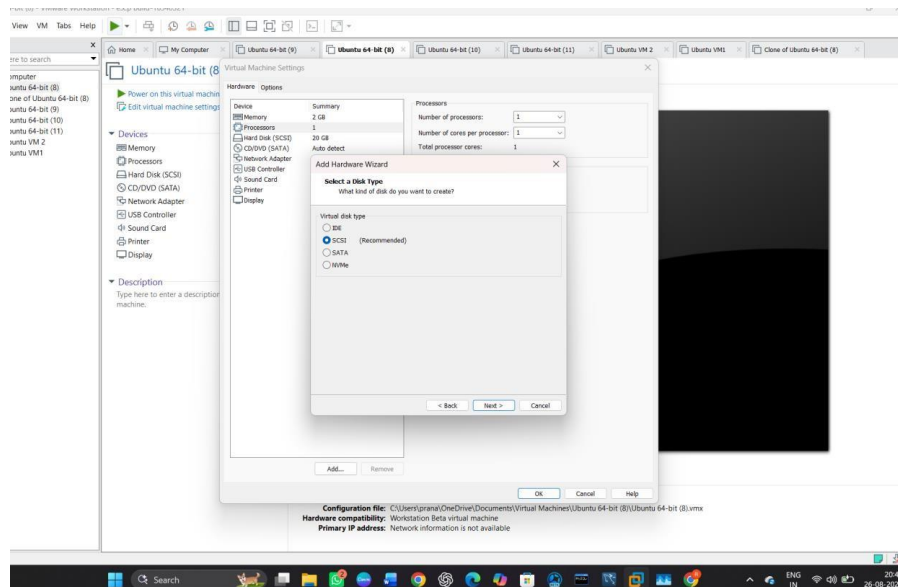
STEP 3 — Open Virtual Machine Settings



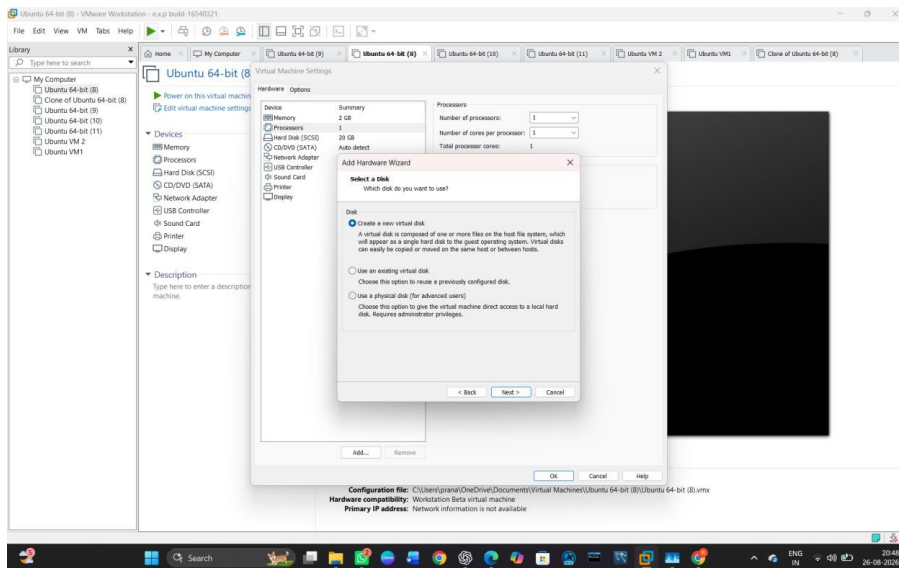
STEP 4 — Add a New Virtual Hard Disk



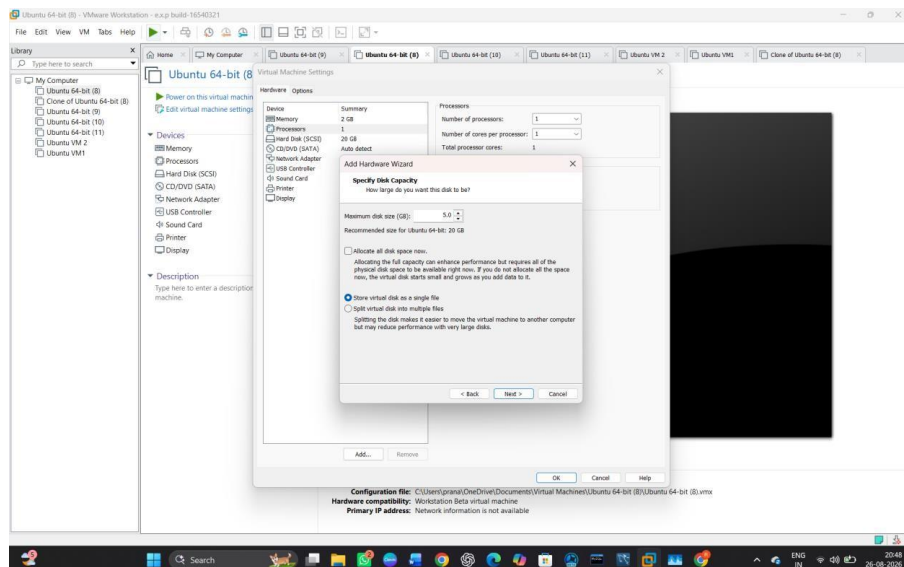
STEP 5 — Select Disk Type



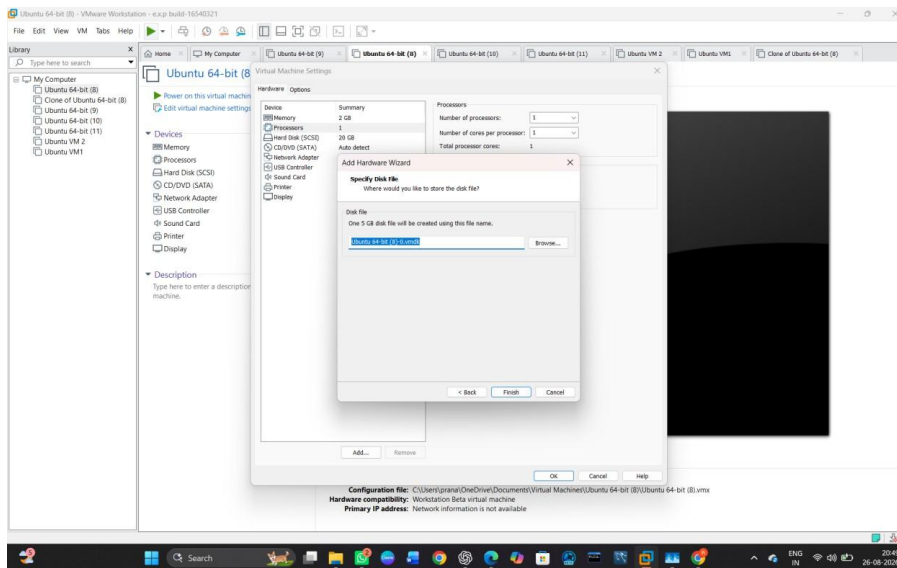
STEP 6 — Create a New Virtual Disk



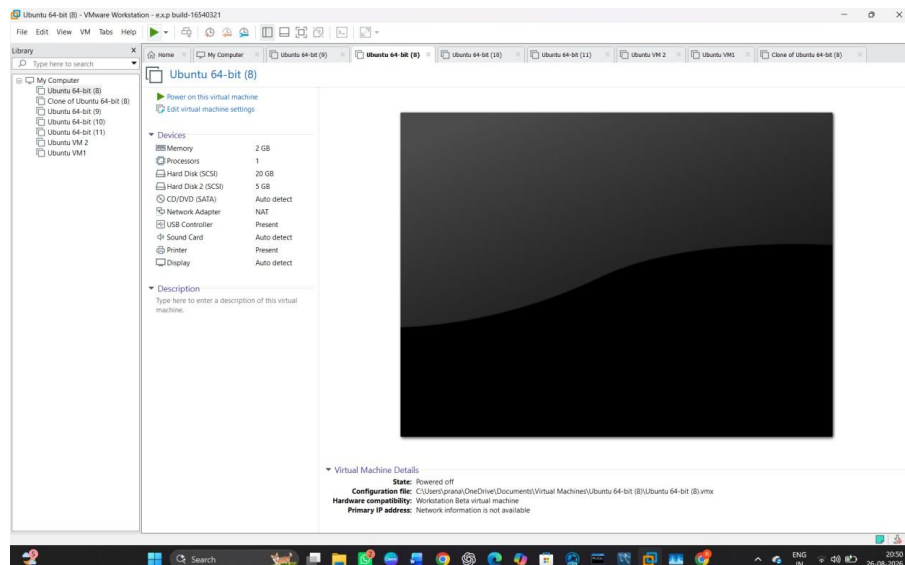
STEP 7— Set Disk Capacity



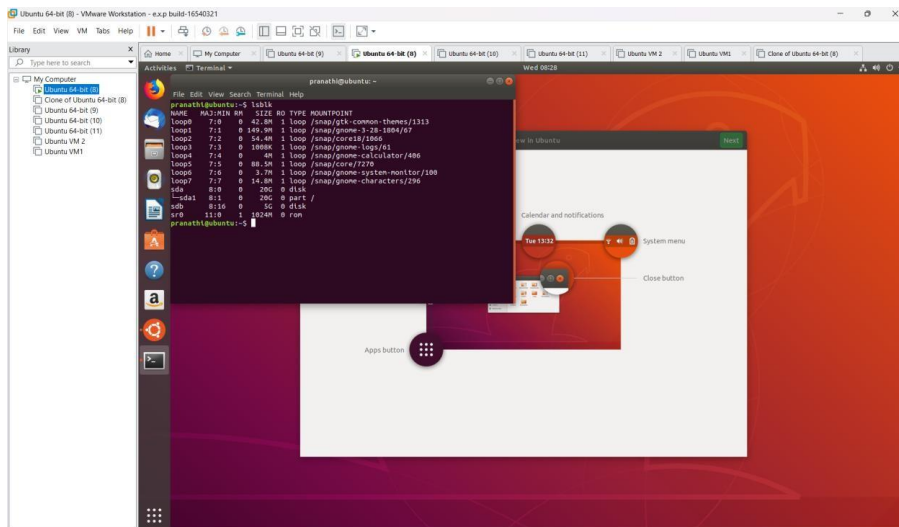
STEP 8 — Specify the Disk File



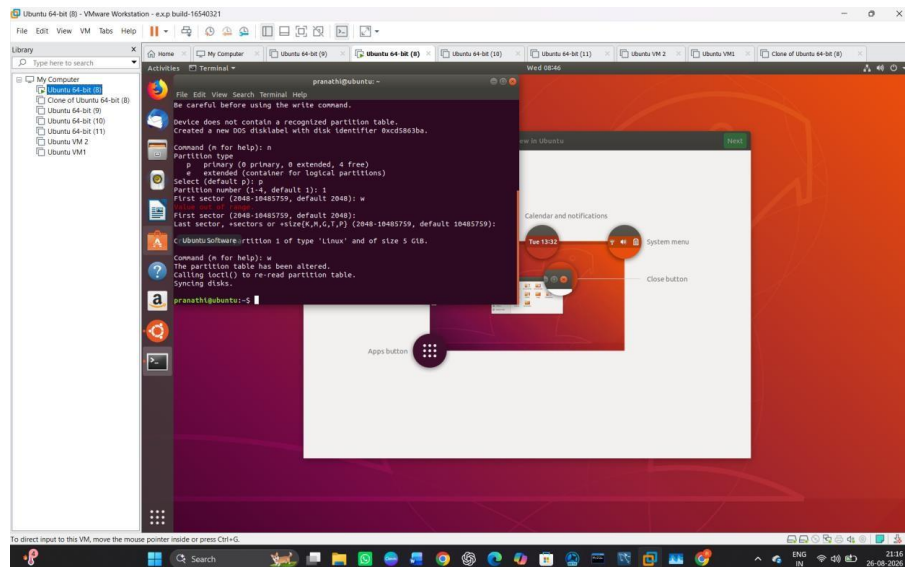
STEP 9 — Confirm the New Disk



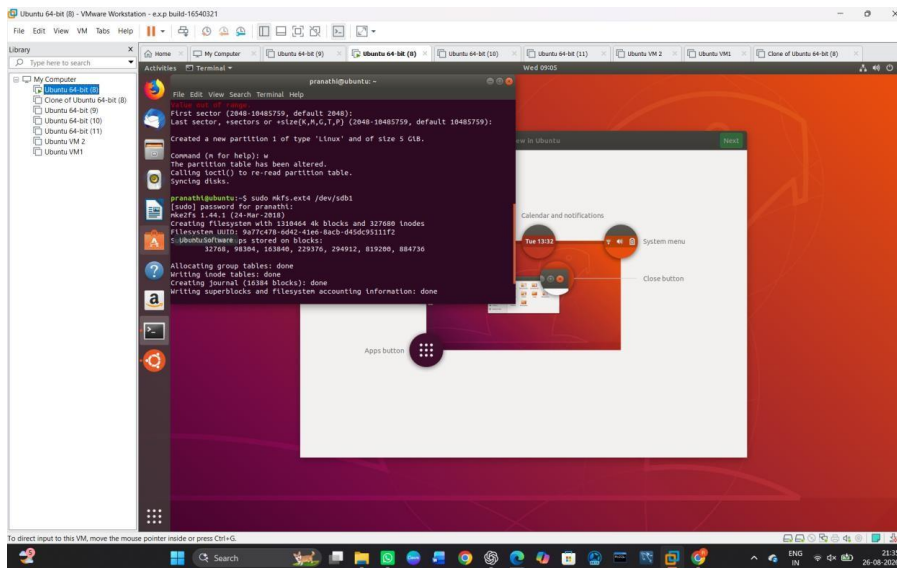
STEP 10 — Check the New Virtual Disk



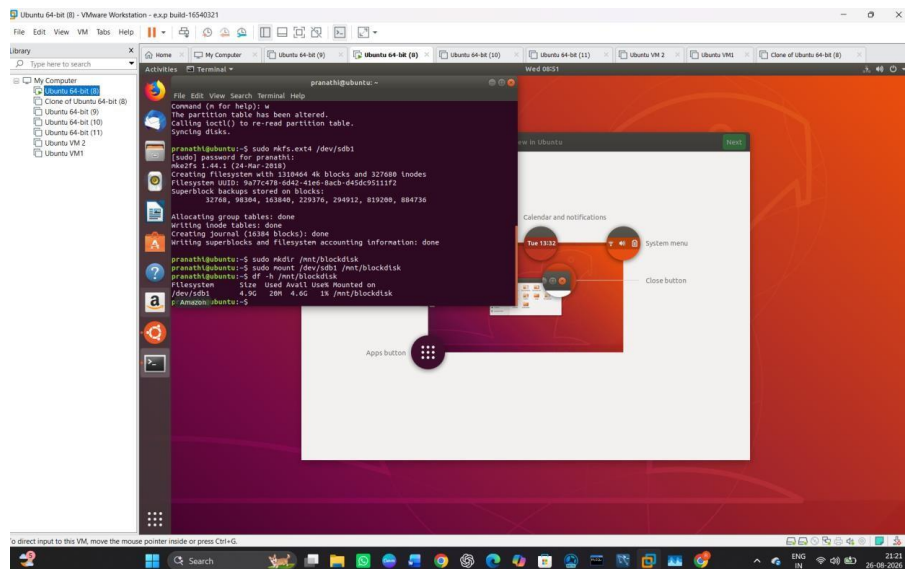
STEP 11 — Create a Partition on the New Disk



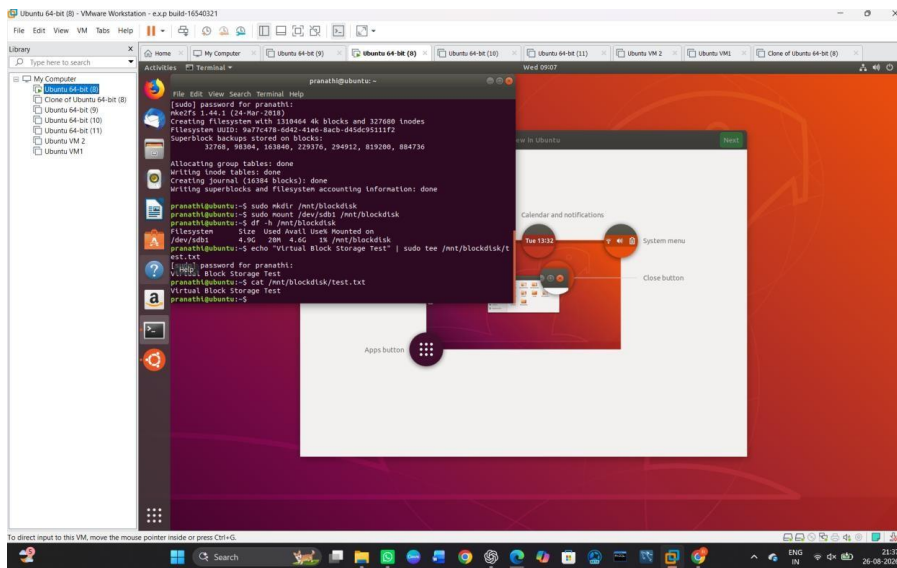
STEP 12 — Format the New Partition



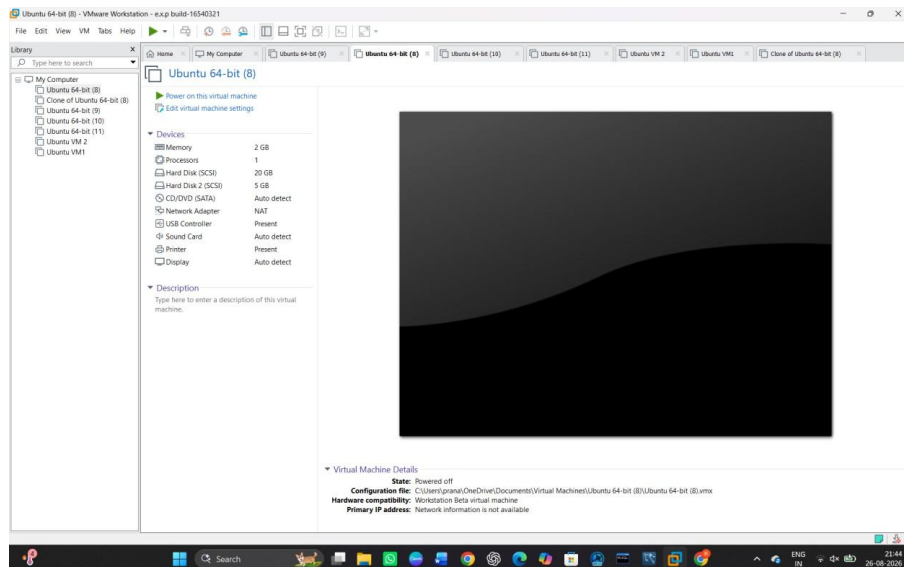
STEP 13 — Create a Mount Point



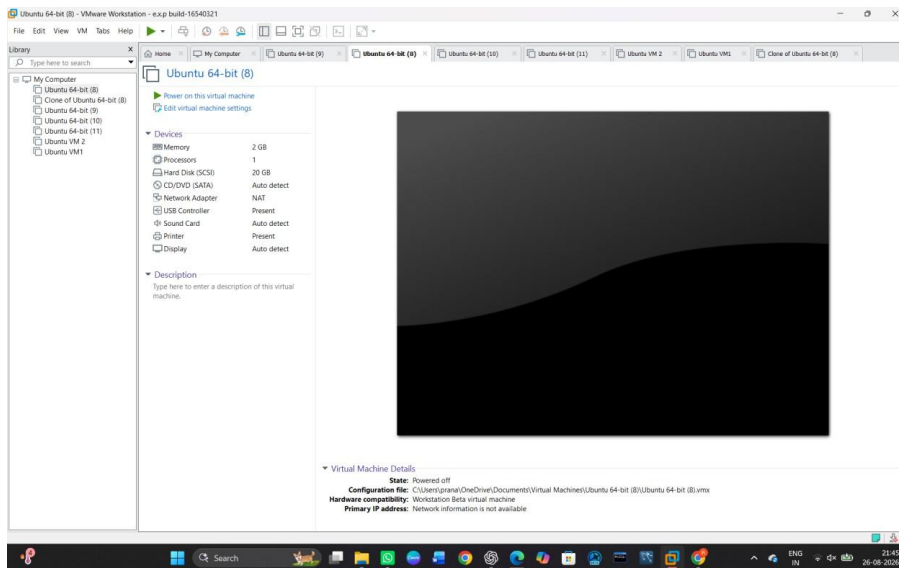
STEP 14 — Create Test Data



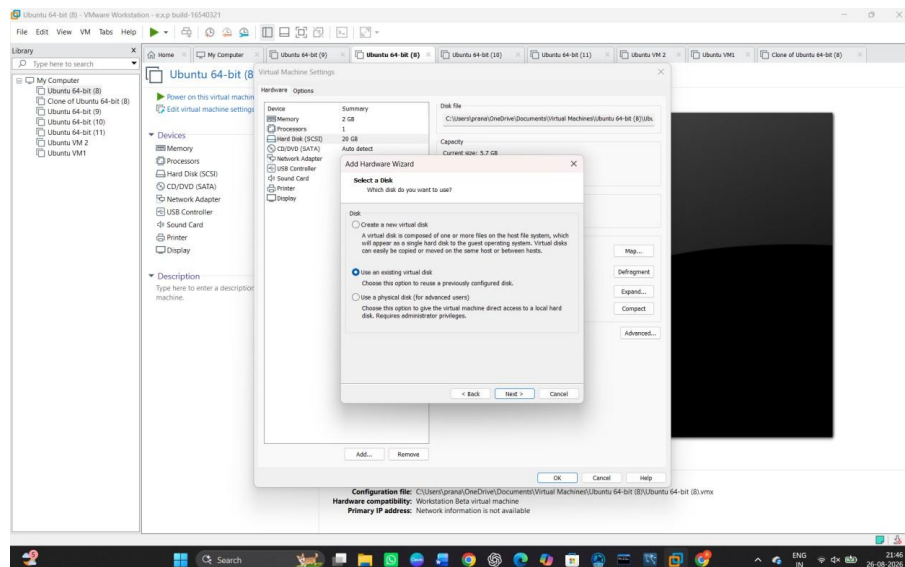
STEP 15 — Shut Down the Virtual Machine



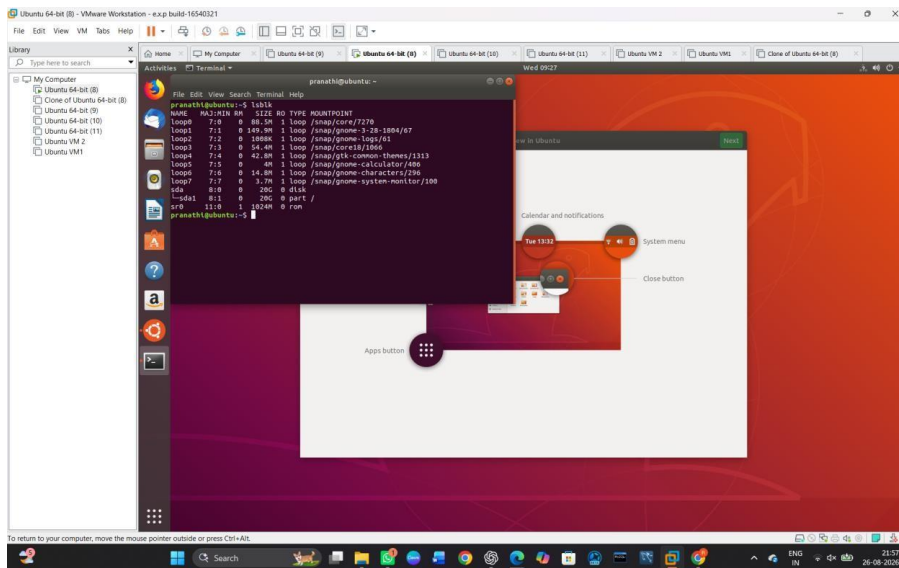
STEP 16 — Detach the Virtual Disk



STEP 17 — Attach the Existing Virtual Disk



STEP 18 — Start Ubuntu Again



Result

The virtual block disk was successfully attached to the virtual machine. Data was stored on the disk, the disk was detached and reattached, and the previously stored data was successfully recovered. Hence, the virtual block storage retains data even after the release of the virtual machine.

Experiment 24:

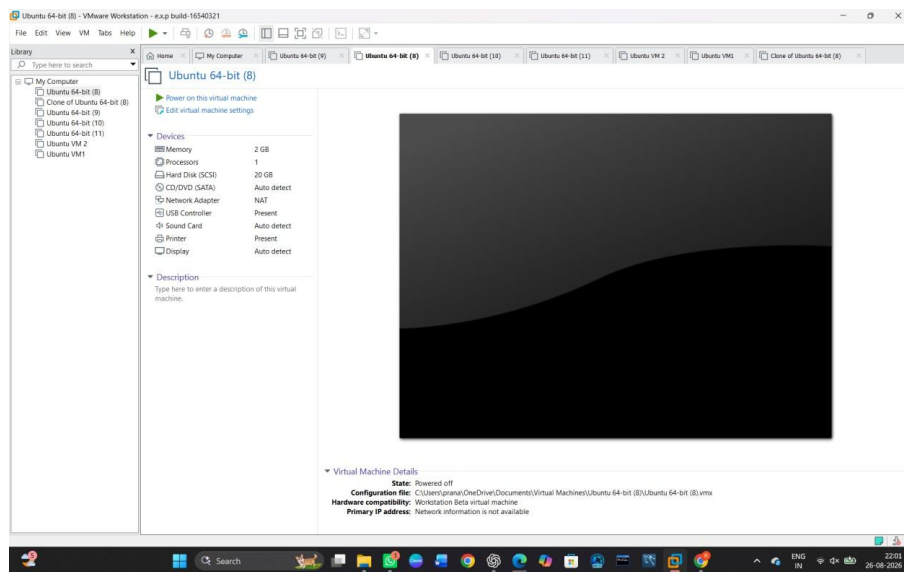
To showcase the virtual machine migration based on the certain condition from one host to the other.

AIM:

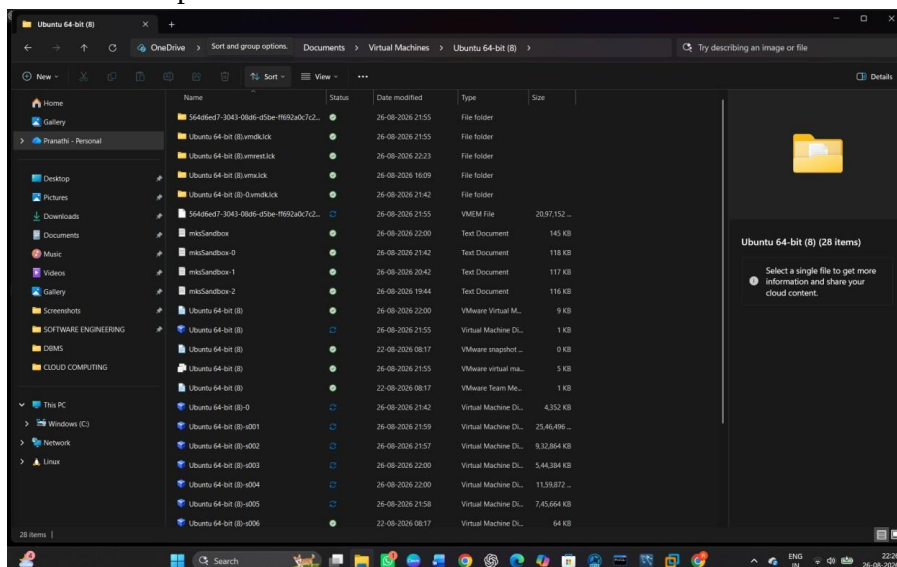
To showcase virtual machine migration from one host to another based on a certain condition.

Procedure:

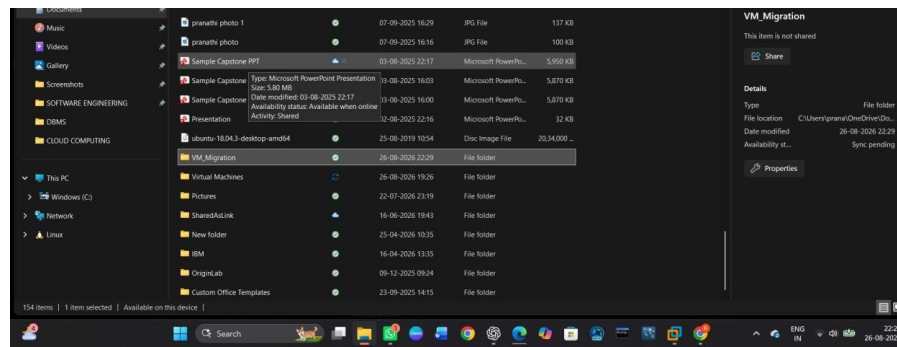
STEP 1 — Prepare the Source VM



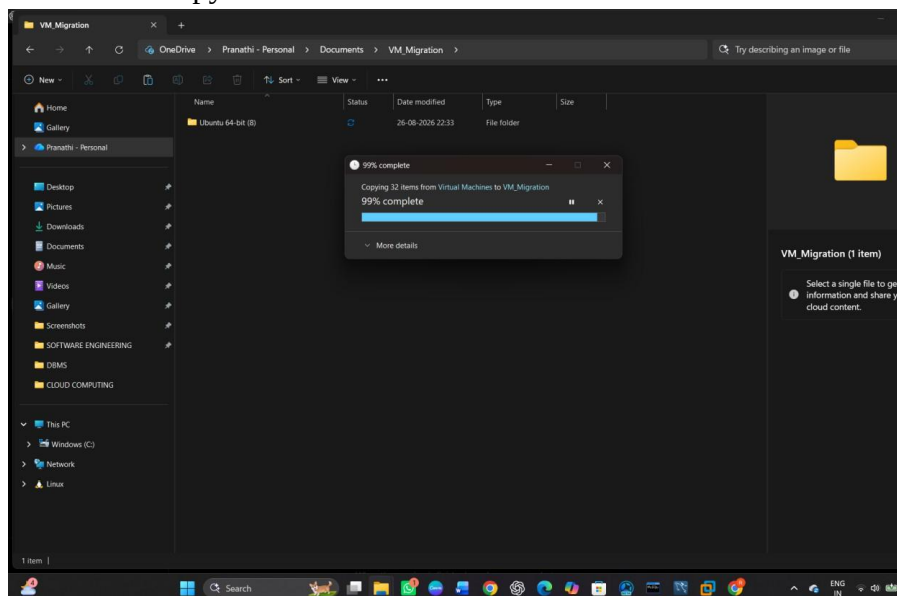
STEP 2 — Open the VM Folder



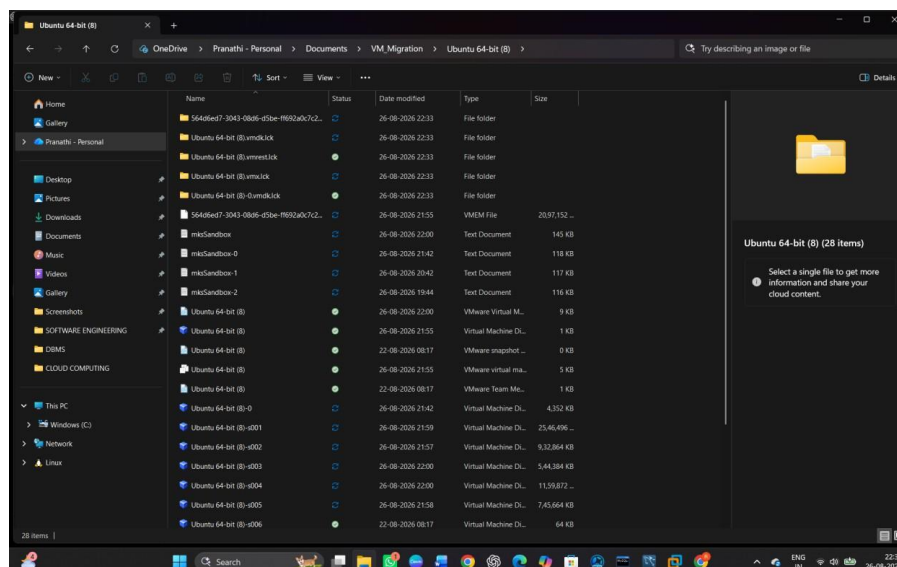
STEP 3 — Create the Destination Folder



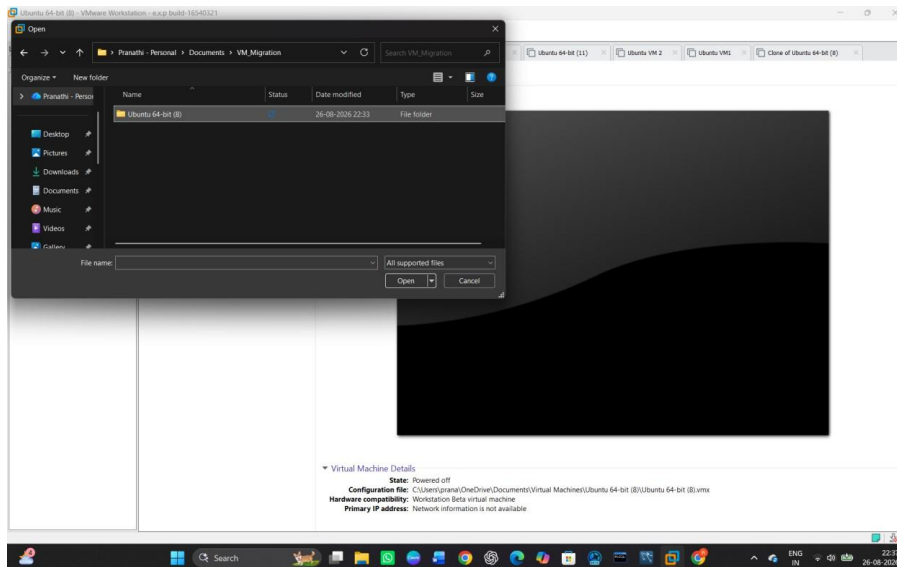
STEP 4 — Copy the Ubuntu VM



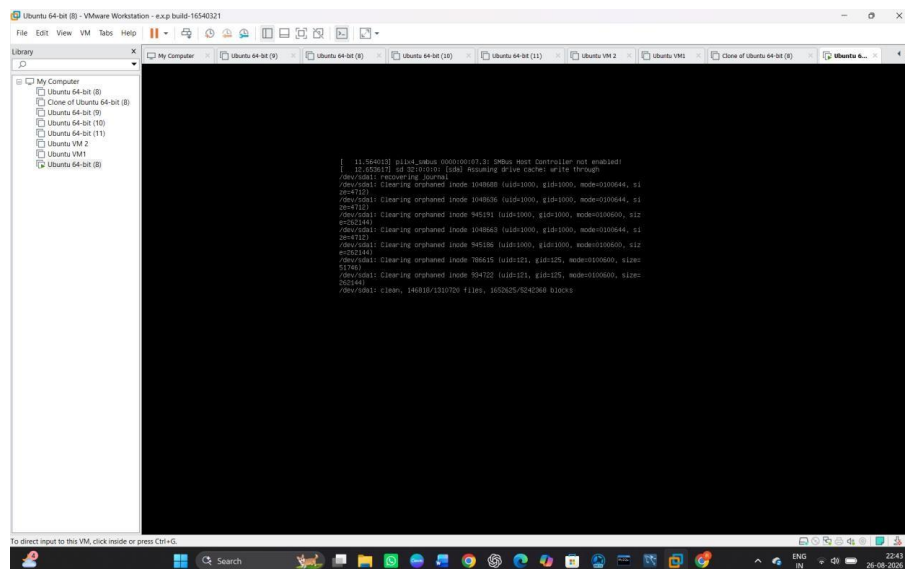
STEP 5 — Open the Copied VM Folder



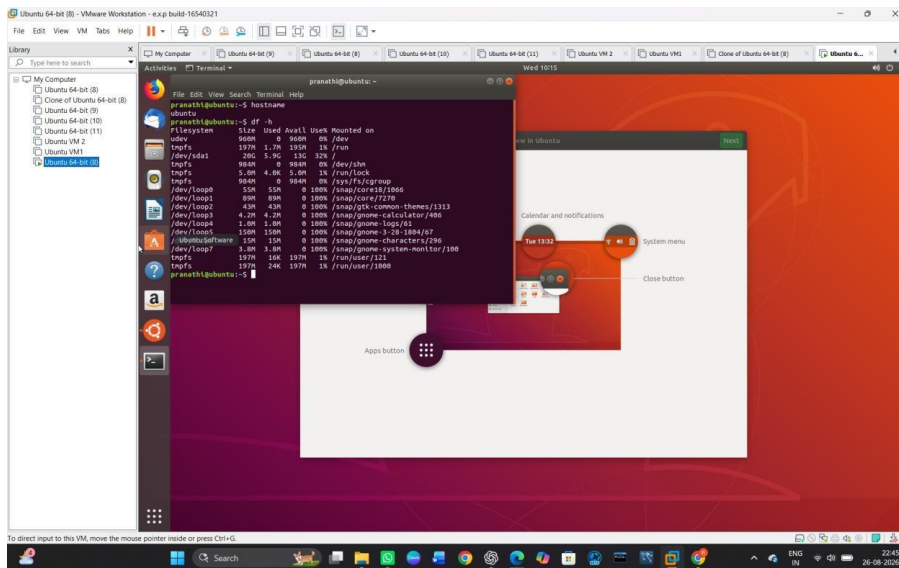
STEP 6 — Load the migrated VM in VMware Workstation.



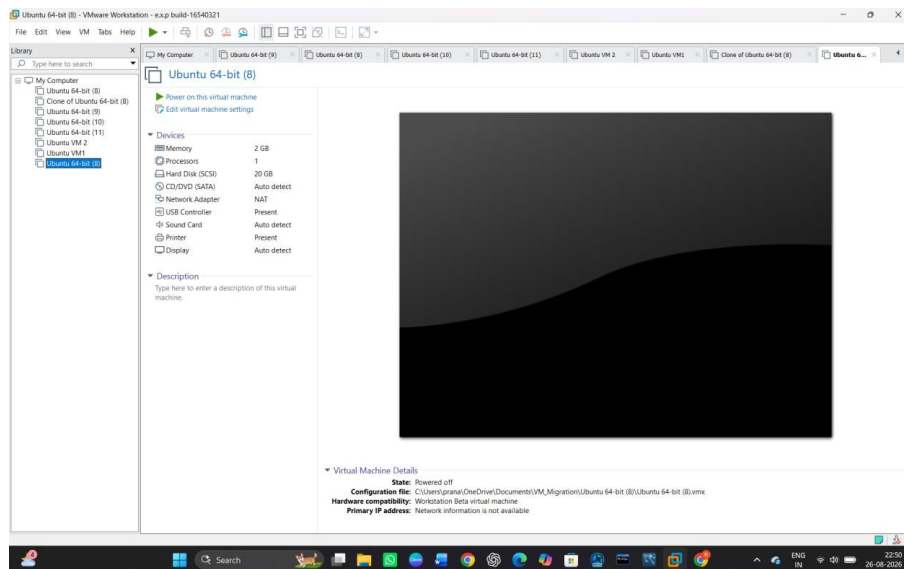
STEP 7 — Start the Migrated VM



STEP 8 — Verify the Migration



STEP 9 — Shut Down the Migrated VM



STEP 10 — Final Verification

